

## Chapter 124 — Bladder Surgery for Benign Disease

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- **Benign structural and functional bladder disorders** — bladder diverticula, distal ureteral strictures, neurogenic bladder, patent urachus and bladder stones — can impair **quality of life** substantially.
- These conditions demand **significant preoperative evaluation and planning** before surgery, both to confirm the diagnosis and to ensure that the chosen operation will succeed.
- Correction draws on **endoscopic, open, laparoscopic and robotic techniques**, often in combination; the emphasis throughout is on patient evaluation, indications, postoperative complications and outcomes.

### BOX 124.1 OPEN AND MINIMALLY INVASIVE SURGERY OF THE BLADDER AND DISTAL URETER

- Bladder diverticulectomy
- Ureteral reimplantation
- Psoas hitch, Boari flap and bladder advancement flaps
- Continent catheterisable channels
- Enterocystoplasty
- Simple cystectomy
- Partial cystectomy and urachal surgery
- Bladder stone and foreign-body removal

- **What the box is for:** it is the operative inventory of the chapter — eight procedures ranging from **bladder-preserving repair** (diverticulectomy, partial cystectomy) through **reconstruction** (reimplantation, flaps, augmentation, channels) to **bladder removal** (simple cystectomy).
- **Pattern across the list:** almost every operation ends the same way — **watertight two-layer closure, catheter drainage for about a week, a pelvic drain, and a cystogram before catheter removal** — so the procedures differ in access and reconstruction, not in aftercare.

## BLADDER ANATOMY AND SURGICAL CONSIDERATIONS

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### *Structure and relations*

- The bladder is a **primarily extraperitoneal hollow organ** in the anterior compartment of the pelvis, composed of an inner **mucosa**, the **lamina propria** and layers of smooth **detrusor muscle**, and surrounded by **perivesical fat**.
- **Anteriorly** it is bordered by the pubic bone and the loose areolar tissue of the **retropubic space (space of Retzius)**. Dissection of this space gives access to the anterior bladder wall and is also used in operations on the prostate and proximal urethra.
- The **transversalis fascia** separates the bladder from the musculature of the anterior abdominal wall.
- **Posteriorly** the bladder is attached to peritoneum, which carries the **umbilical ligaments** upward to the umbilicus:
  - the **medial umbilical ligaments** contain the **obliterated umbilical arteries**, branches of the internal iliac artery;
  - the **median umbilical ligament** is the **urachus**, the fibrous remnant of the **allantois**. Developmental failure of the urachus produces a **patent urachus, urachal sinus or urachal cyst**.

- The **peritoneal reflection** lies posteroinferiorly — near the **prostatorectal junction** in men and in the **vesicouterine space** in women. Incising the reflection here opens the space behind the bladder and trigone.
- Directly behind the bladder lies the **vagina** in women; in men **Denonvilliers fascia** separates the bladder and prostate from the rectum.
- **Laterally** the bladder abuts the pelvic sidewall, including the distal obturator space. Laterally and posteriorly the **round ligament** (women) or **vas deferens** (men) is closely related.

#### *Embryology and the ureter*

- The bladder derives from the **urogenital sinus (endoderm)**; its detrusor consists of longitudinal layers of smooth muscle. Developmental defects can produce structural anomalies of the urethra, bladder and ureteric insertion.
- After crossing the common iliac vessels the ureters run **lateral to medial and cranial to caudal**.
- The two classic crossings form an examinable pair:

| Sex    | Structure crossing over the ureter               | Site                                |
|--------|--------------------------------------------------|-------------------------------------|
| Female | <b>Uterine artery</b> ("water under the bridge") | Near the internal os of the cervix  |
| Male   | <b>Vas deferens</b>                              | Near the tip of the seminal vesicle |

- The distal ureter passes **medial to the medial umbilical ligament** and inserts at the **trigone**, at the bladder base on the posteroinferior wall.

#### *Blood supply, lymphatics and innervation*

- Arterial supply is chiefly from the **internal iliac artery**:
  - the **superior vesical artery** arises from the **umbilical artery**;
  - the **inferior vesical artery** arises directly from the internal iliac (from the **vaginal artery** in women).
- The superior and inferior vesical arteries form the **bladder pedicles**, which enter **posterolaterally**. **Partial division of one pedicle** aids bladder mobilisation.
- Venous drainage is via a **vesical venous plexus** into the internal iliac veins.
- Lymphatic drainage is mainly to the **external iliac nodes**, with additional drainage to the obturator, internal iliac and common iliac nodes.
- Innervation: **sympathetic T10–L2** (thoracolumbar) and **parasympathetic S2–S4** (sacral). Somatic supply to the urethral sphincter is via the **pudendal nerve**, whose motor neurons lie in **Onuf's nucleus** in the ventral horn at S2–S4.

**Concept** — Nearly every benign bladder operation is organised around three landmarks: the **space of Retzius** for access, the **peritoneal reflection** for exposure of the posterior bladder, and the **distal ureter** for safety; knowing where each lies decides whether an operation can stay extraperitoneal.

## BLADDER DIVERTICULECTOMY

### Description

- A bladder diverticulum is an **outpouching of the bladder wall devoid of a functional muscularis propria**. Its wall consists of **mucosa and lamina propria**, with only **scattered, architecturally disorganised detrusor fibres** that cannot contract in a coordinated way. The result is **urinary stasis** and its sequelae, which prompt symptoms and intervention.

- Outside the diverticulum lies the adventitia of the bladder wall, around which a **fibrous capsule (pseudocapsule)** may form; this capsule provides the **surgical cleavage plane** when the diverticulum is resected.
- **Acquired diverticula (adults)** arise most often from **bladder outlet obstruction (BOO)** and the resulting high voiding pressure.
  - There is a **male predominance** because **benign prostatic hyperplasia (BPH)** causes the majority; urethral stricture can also cause them, and in women pelvic organ prolapse may produce outlet obstruction.
  - **Neurogenic bladder with poor wall compliance** raises intravesical pressure and also produces diverticula.
  - Diverticula formed in these settings are frequently **multiple and of varying size** — factors that strongly influence both the need for intervention and the choice of treatment.
- **Most acquired diverticula are asymptomatic** and are found incidentally on imaging or cystoscopy performed for another reason, often for the underlying condition.
- **Symptoms arise from stasis: urinary tract infection (UTI) and bladder calculi**; a large diverticulum may also hold enough urine to cause a sensation of incomplete emptying. Less often a large diverticulum **compresses the ureter extrinsically** near its insertion, causing ureteral obstruction and **hydronephrosis**.
- An **increased risk of bladder cancer within diverticula** has been observed, so a thorough evaluation is mandatory before any intervention.
- **Congenital diverticula** are commoner in children and arise from an **intrinsic weakness of the bladder musculature**, not from outlet obstruction or neurogenic disease. Nevertheless a substantial proportion of these patients have voiding dysfunction, so they too require evaluation for outlet obstruction and neurogenic dysfunction.
- Congenital diverticula are usually **solitary** and **posterolateral, close to the ureteric orifice** — the **Hutch diverticulum**. Because the herniation disturbs the detrusor at the ureteric hiatus, **vesicoureteral reflux (VUR) is generally present** and may cause hydronephrosis or pyelonephritis; **surgical planning should therefore include ureteral reimplantation**.
- **Less common causes in adults**: paediatric neurogenic and obstructive conditions (posterior urethral valves, prune-belly syndrome); genetic connective tissue disorders (Ehlers-Danlos and Williams syndromes); and **iatrogenic disruption of the bladder**, most often in women after anti-incontinence procedures.

#### ACQUIRED VERSUS CONGENITAL BLADDER DIVERTICULA

| Feature              | Acquired                                                                                                            | Congenital                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Typical population   | Adults, <b>male predominance</b>                                                                                    | <b>Children</b>                                                            |
| Mechanism            | <b>High voiding or storage pressure</b> — outlet obstruction (BPH commonest) or poorly compliant neurogenic bladder | <b>Intrinsic weakness of the detrusor</b>                                  |
| Number and size      | Often <b>multiple</b> , variable size                                                                               | Usually <b>solitary</b>                                                    |
| Site                 | Variable; associated with trabeculation                                                                             | <b>Posterolateral, beside the ureteric orifice (Hutch)</b>                 |
| Associated finding   | Trabeculation, calculi, UTI                                                                                         | <b>VUR generally present</b>                                               |
| Surgical implication | <b>Treat the outlet or compliance problem</b> before or with the diverticulum                                       | <b>Plan ureteral reimplantation</b> ; still screen for voiding dysfunction |

## Evaluation and Surgical Indications

- **Cystoscopy is central to the work-up.** It allows direct inspection of the diverticulum and, above all, **evaluation for malignancy**, which would change the treatment entirely. Record:
  - the **proximity of the diverticular mouth to the ureteric orifices**;
  - the **number** of diverticula, the **size of the mouth**, and an **approximate volume**.
- **Every diverticulum should be entered where possible and its wall inspected**, because of the reported cancer risk. Stones formed by stasis can be treated at the same time as the diverticulectomy.
- **Trabeculation** points to outlet obstruction as the cause. An **intravesical prostatic lobe**, urethral stricture or bladder neck contracture should be noted, because each will need treatment **before or at the time of** diverticulum surgery.
- **Voiding cystourethrogram (VCUG)** defines location, size and number; shows **VUR** and other anatomical abnormalities; and reveals diverticula **missed at cystoscopy**, particularly amid heavy trabeculation.
  - It reliably shows the **relationship of the diverticulum to the ureter and any reflux**, which decides whether a **preoperative ureteral stent** (for identification and protection) or a **simultaneous ureteral reimplantation** is needed.
  - It also shows **bladder neck obstruction** — bladder neck hypertrophy, functional bladder neck obstruction, detrusor–bladder neck dyssynergia, posterior urethral valves or urethral stricture.
- **Upper tract imaging:**
  - **Renal ultrasound** detects hydronephrosis but resolves the ureter and calculi poorly.
  - **Triphasic computed tomography (CT) of the abdomen and pelvis with a delayed excretory phase (CT urogram)** shows hydronephrosis, the **course of the ureter** and any stones. Hydronephrosis may result from **direct compression** by the diverticulum or from **VUR through a weakened detrusor tunnel**.
  - Cross-sectional imaging (CT or **magnetic resonance imaging, MRI**) also resolves the number, size and location of diverticula; excretory images can show **lateral displacement of the ureter** by the diverticulum.
- **If contrast cannot be given:** ultrasound or non-contrast CT for hydronephrosis, combined with **intraoperative retrograde pyelography and cystography** to define the diverticulum–ureter relationship and ureteral obstruction. **Magnetic resonance urography** may also be considered.
- **Characterise the underlying cause**, since it must be treated before or with the diverticulum: in-office **flow rate** or formal **urodynamics** for detrusor contractility and voiding pressure. Where BPH will be treated concurrently, estimate **prostate volume** by **transrectal ultrasonography (TRUS)**, CT, MRI or cystoscopy.
- **Urine cytology in all patients**, because of the cancer association.
- **A negative urine culture before any surgery**; treat any preoperative UTI with a culture-directed antibiotic course.

## INDICATIONS FOR TREATMENT OF BLADDER DIVERTICULA

| Clinical situation                                                                                        | Management                                                          |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Small, asymptomatic, uncomplicated                                                                        | <b>Observation</b> , or cystoscopic <b>fulguration or resection</b> |
| Large with <b>incomplete emptying</b> , chronic or recurrent <b>UTI</b> , <b>calculi</b> , or <b>pain</b> | <b>Diverticulectomy</b>                                             |
| Causing <b>VUR</b> or <b>ureteral obstruction</b>                                                         | Diverticulectomy, <b>often with ureteral reimplantation</b>         |

| Clinical situation                 | Management                    |
|------------------------------------|-------------------------------|
| Malignancy within the diverticulum | Partial or radical cystectomy |

- **The table's trap:** a diverticulum harbouring cancer is not a diverticulectomy case at all — it becomes a cystectomy decision.

**Concept** — A diverticulum is a **marker of the bladder behind it**: the operation succeeds only if the outlet or compliance problem that created it is treated, and only after cancer and ureteric involvement have been excluded.

## Surgical Technique

- **History:** diverticulectomy was first described in **1897**, and treatment has evolved from open surgery to endoscopic, laparoscopic and robotic techniques.
  - **Transperitoneal laparoscopic bladder diverticulectomy (LBD)** was introduced in **1992**; subsequent reports confirmed its reproducibility by both **transvesical** and **extravesical** routes.
  - **Robotic bladder diverticulectomy (RBD)** was described in **2007**. It mimics the laparoscopic technique, helps the surgeon who lacks extensive laparoscopic experience, and gives a comfortable ergonomic platform for work in the deep pelvis.
- **Peritoneal access is the governing contrast between open and minimally invasive surgery:**

|             | Extraperitoneal                                                                                     | Transperitoneal                                                                            |
|-------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Usual route | <b>Open surgery</b> — possible even for posterior diverticula                                       | <b>Laparoscopic and robotic surgery</b>                                                    |
| Advantage   | <b>Less risk of visceral injury and of intraperitoneal urine leak</b> (a major source of morbidity) | <b>Wide, generous working space</b>                                                        |
| Evidence    | —                                                                                                   | Fewer outcome reports exist for extravesical/extraperitoneal minimally invasive approaches |

- **Choice of technique depends on** the number and location of the diverticula, their **proximity to the ureter**, and the need for **concomitant ureteral or outlet surgery**.
- **Complex (large, posterior) and multiple diverticula needing ureteral reimplantation** have traditionally been managed **open**, although laparoscopic or robotic surgery is reasonable in sufficiently experienced hands.
- **Alternative minimally invasive approaches:**
  - **Transvesical pneumovesicoscopic diverticulectomy** — carbon dioxide pneumovesicum established cystoscopically, then trocars placed through the abdominal wall directly into the bladder.
  - **Laparoendoscopic single-site surgery (LESS)** — by transabdominal and by transvesical routes.
  - **da Vinci single-port (SP) robot** — extravesical and transvesical diverticulectomy for posterior diverticula, shown to be technically feasible.
- **The underlying cause must be addressed before or at the time of diverticulectomy** — relief of outlet obstruction, or **augmentation cystoplasty for neurogenic bladder**.
  - When BPH is the cause, prostate surgery can be done in the same sitting, **immediately before or after** the resection: transurethral endoscopic procedures, or extirpative surgery such as **simple prostatectomy, particularly for glands larger than 80 cc**. Laparoscopic or robotic simple prostatectomy can accompany LBD or RBD.
  - **If the outlet is treated endoscopically, do it before the diverticulectomy** — otherwise irrigation fluid distends the bladder and tensions fresh suture lines, risking **perforation**.

## Open Technique

### *Intravesical approach*

- **Positioning:** supine with the **anterior superior iliac spine (ASIS) over the break of the table**, and the table slightly flexed to open the space of Retzius. The genitalia and infraumbilical abdomen are prepared and draped.
- **Bladder preparation:** an **18 Fr Foley catheter** is placed on the sterile field, the bladder is filled with **about 200 mL of sterile water**, and the catheter is plugged.
  - *Alternative:* flexible cystoscopy at the start; a guidewire is used to place a **Councill-tip catheter into the diverticulum**, so that the diverticulum can be filled selectively for intraoperative localisation (while a Foley catheter separately fills the bladder).
  - A **ureteral catheter** is placed when the diverticulum lies close to the ureter.
- **Access:** lower midline incision to the pubic symphysis → subcutaneous tissue divided with electrocautery → rectus fascia opened along the **linea alba** → rectus bellies split → **transversalis fascia** incised, exposing preperitoneal fat above and the space of Retzius below.
  - **Bladder distension** pushes the **peritoneal reflection cephalad**, easing entry into the space of Retzius; the bladder can be distended further for more displacement.
  - A sponge-stick develops the space of Retzius and the **lateral pelvic gutters** bluntly, pushing perivesical fat medially and sweeping the peritoneal reflection off the dome to avoid inadvertent peritoneal entry. A self-retaining retractor is placed.
- **Cystotomy:** vertical, in the midline, from the anterior dome to **just above the bladder neck**, aspirating the fluid as it lengthens. **2-0 polyglactin stay sutures at the apices** prevent tearing. All fluid is aspirated and the **mucosa inspected for suspicious lesions**; deeper retractor blades (or a hand-held deep blade) expose the bladder walls.
- **Identify the ureteric orifices** and note their relation to the diverticulum; insert a **5 Fr ureteral catheter** if there is concern.
- **If the diverticulum can be everted:** an **Allis clamp** is passed through the neck to grasp its fundus and evert it into the bladder; the mucosa is incised circumferentially around the neck with a **margin of about 0.5 cm**, carried through the submucosal connective tissue to free the sac from the bladder wall.
- **If it cannot be everted:** a circumferential mucosal incision is made **about 0.5 cm from the mouth** into the submucosal connective tissue. The **plane between the diverticular sac and its fibrous pseudocapsule** is developed by blunt and sharp dissection, with an Allis clamp on the neck — repositioned around its perimeter — providing countertraction, until the sac is **freed and removed entire**. **No drain is needed in the remnant capsule.**
- **Closure of the defect:** two layers of running **2-0 polyglactin**, with a **secure seromuscular closure to minimise recurrence**.
- The catheter is exchanged for a **larger Foley (20–22 Fr)**; the cystotomy is closed in two running layers of 2-0 polyglactin — **mucosa first, then seromuscular**.
- **Leak test:** the bladder is gently filled with sterile water; any leak is reinforced with **figure-of-eight 2-0 polyglactin**.
- **Suprapubic catheter** — **optional**, at the surgeon's discretion to maximise drainage and reduce tension on the suture lines: a separate punch cystotomy lateral to the midline cystotomy, the catheter threaded in under vision and held with a **2-0 polyglactin purse-string**, then brought out through a stab incision **lateral to the rectus belly** and secured with nonabsorbable suture.
- **Pelvic drain:** ideally **anterior to the cystotomy, curving into the lateral gutter beside the diverticulectomy site**; out through a separate stab, secured with nonabsorbable suture, on **self-suction**.

- Fascia closed (continuous or interrupted); subcutaneous tissue irrigated; skin closed.

#### *Combined intravesical and extravesical approach*

- The opening steps match the intravesical technique: cystotomy, identification of the mouth, and a circumferential mucosal incision **about 0.5 cm from the mouth** into the submucosal connective tissue.
- A **finger or blunt curved instrument (for example a metal suction tip) is placed inside the sac** to apply anterior traction, bringing its extravesical portion into view. A **Kittner sponge** teases perivesical fat off the external surface, exposing the junction of sac and bladder.
- With outward traction on the sac, the **extravesical surface of the neck is incised circumferentially** with electrocautery.
- **The extravesical incision must meet the intravesical mucosal incision.** Failure to establish this continuity creates a new plane and leads to **resection of excess bladder wall**.
- The detached sac, still inside its pseudocapsule (which may adhere to perivesical tissue), is dissected free bluntly and sharply with Allis countertraction at the neck. The defect is closed **intravesically or extravesically** in two running layers of 2-0 polyglactin.

#### INTRAVESICAL VERSUS COMBINED INTRAVESICAL-EXTRAVESICAL OPEN DIVERTICULECTOMY

|                          | Intravesical                                                                  | Combined intravesical-extravesical                                      |
|--------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| How the sac is delivered | <b>Everted into the bladder</b> , or dissected out of its capsule from inside | Pushed <b>outward by a finger in the sac</b> and dissected from outside |
| Plane used               | Sac-pseudocapsule plane from within                                           | Extravesical neck, then sac-pseudocapsule plane                         |
| Specific hazard          | Ureteric orifice injury                                                       | <b>Excess bladder wall resection</b> if the two incisions do not meet   |
| Defect closure           | Intravesical, two layers                                                      | Intravesical or extravesical, two layers                                |

#### Laparoscopic and Robotic Techniques

- **Positioning: dorsal lithotomy**, arms secured at the sides, pressure points padded ("egg-crate" foam), patient taped to the table, optionally on a non-skid pad; then **steep Trendelenburg**. The abdomen and groin are prepared.
- **Cystoscopy before port placement:**
  - **ureteral catheterisation** and retrograde pyelography when needed;
  - **selective catheterisation of the diverticulum** with a **Councill-tip catheter over a guidewire**, plus a separate Foley catheter in the bladder, for selective filling and identification;
  - or **real-time transillumination** of the diverticulum with a flexible cystoscope — particularly valuable with **multiple diverticula**, where catheterising each one is impractical;
  - **intravesical methylene blue** has been used to identify the bladder neck during robotic surgery.
- **Access (transperitoneal LBD and RBD):** Veress needle through a periumbilical incision; **pneumoperitoneum at 15 mm Hg**. In patients with previous abdominal surgery a **subcostal Veress site** can be used. Steep Trendelenburg displaces the bowel cephalad.
- **Robotic port placement:**
  - **Camera port** in the midline **about 2 cm above the umbilicus** — **12 mm for the Si system, 8 mm for the Xi system**; the abdomen is inspected for injury and the remaining ports placed under vision.
  - Two **8-mm robotic ports** on either side, caudal to the camera at umbilical level, **about 10 cm lateral** to it near the lateral border of the rectus.

- Optional **fourth arm (8 mm)** about 8 cm lateral to one pararectus port and **two fingerbreadths above the ASIS, ideally on the side of the diverticulum**.
- **12-mm assistant port contralateral to the diverticulum**, two fingerbreadths above the ASIS — this gives the assistant direct access, whereas an ipsilateral port is restricted by the iliac bone.
- Optional **5-mm assistant port** in the upper abdomen contralaterally, triangulated about 8 cm from the camera and paramedian port.
- The robot is **docked at the side for Xi, between the legs for Si**.
- **Laparoscopic port placement: 10-mm umbilical camera port**; for a right-handed surgeon, a **12-mm right** and **5-mm left pararectus** working port; an optional 5-mm port about 2 cm cephalad and medial to the ASIS.
- **Dissection sequence:**
  - incise the peritoneum over the diverticulum, located by **selective filling or cystoscopic transillumination** (the cystoscope is then moved to the chosen entry point anterior to the neck, leaving a margin) →
  - for **large or multiple diverticula**, mobilise the bladder: incise peritoneum **lateral to the obliterated umbilical ligaments bilaterally**, divide the **urachus** and "**drop**" the bladder into the space of Retzius →
  - energy: **10-mm LigaSure for LBD; monopolar and bipolar instruments for RBD** →
  - divide peridiverticular adhesions down to the neck →
  - circumscribe the extravesical neck (**monopolar hook for LBD, monopolar scissors for RBD**), beginning anteriorly, with traction on the sac (**ProGrasp forceps robotically, atraumatic bowel graspers laparoscopically**) →
  - carry the incision through the bladder wall to excise the diverticulum completely →
  - place the specimen in an **Endo Catch bag**.
- **Constant vigilance protects the ureter and its orifice**; the ureter may need careful dissection off the external surface of the sac.
- **Transvesical variant (selected cases)**: use cautiously if there is **ureteric involvement or a planned reimplant**. After dropping the bladder, open it in the midline; **clamp the Foley catheter externally** to maintain pneumoperitoneum; pull the diverticulum into the bladder (inversion helped by pushing on its outer surface while pulling on its mucosa from within); circumscribe the mucosa at the neck and deepen the incision to detach it.
- **Closure**: the defect (and, for the transvesical route, the cystotomy) in **two layers of 2-0 polyglactin**; the bladder can be **hitched anteriorly** to expose the defect.
  - Laparoscopically: freehand intracorporeal suturing or the **Endo Stitch** device.
  - An **Endo GIA stapler** can excise the diverticulum, but carries a **theoretical risk of stone formation** on the staples.
  - The bladder is filled to confirm a **watertight closure**.
- **Ureteral reimplantation** is added when needed for reflux, obstruction or iatrogenic injury, and is **often necessary when the diverticulum involves the ureteral hiatus** — the neck is amputated, the ureter separated from the bladder, the diverticular opening closed, a new anterior-wall opening created, a **double-J stent** placed and the ureter sewn to the bladder.
- The peritoneum may be closed to separate intraperitoneal and extraperitoneal compartments. A **closed-suction pelvic drain** is brought out through a trocar site; pneumoperitoneum is released; the specimen is extracted through an **extension of the midline camera-port incision**; the extraction-site and 12-mm port fascia are closed with **figure-of-eight 0-polyglactin**.

- **Extraperitoneal LBD:** incision **2 cm below the umbilicus** through fascia and linea alba → blunt finger dissection, extended with a **dilating balloon** → **Hasson port** → carbon dioxide insufflation of the preperitoneal and retropubic space → two pararectus ports (5 or 12 mm) inferior to the incision, plus an optional midline assistant port between symphysis and umbilicus.
- **LESS diverticulectomy** gives **excellent cosmetic results**. By the transabdominal route it follows the LBD/RBD steps. The **transvesical single-site** version places a single-port device into the bladder under cystoscopic guidance, establishes pneumovesicum, circumscribes the neck with monopolar scissors or a hook, excises the sac with **transurethral/cystoscopic retraction**, closes the defect and extracts the specimen.
- **da Vinci SP robot** (first diverticulectomy reported in **2019**): usable **intraperitoneally**, or by **regionalised extraperitoneal** and **transvesical** routes.
  - *Extraperitoneal:* incision 1–2 cm below the umbilicus → rectus fascia opened, recti split → space of Retzius developed digitally or with a balloon → small wound retractor with a gel-port or dedicated access port.
  - *Transvesical:* **3-cm midline incision about 2 cm above the pubic symphysis** → bladder filled with about 200 mL → bladder confirmed by **aspiration with an 18-gauge needle** → stay sutures, cystotomy, wound retractor inner ring and gel-port → dock and proceed as above.

## Postoperative Care and Complications

- **Foley catheter for about 1 week** to allow the closure to heal.
- **Pelvic drain:** output is monitored — **high output suggests a urinary leak**. If output is low and catheter drainage good, the drain is removed, typically at **24–48 hours**. Some surgeons measure **drain-fluid creatinine** to confirm the integrity of the closure.
- **A cystogram before outpatient catheter removal** confirms no extravasation; periprocedural antibiotics are **optional** to reduce transient bacteraemia.
- Complications are **similar after open and minimally invasive surgery**.
- **Urinary extravasation is the commonest complication**. It may appear only as **high drain output**, or symptomatically as abdominal pain and fever from an **infected urinoma**; after a **transperitoneal** approach it may present with **ileus and electrolyte abnormalities**.
  - Confirm the leak with **drain-fluid creatinine compared with serum creatinine**; characterise its site and degree with **CT cystography**.
  - **Management ladder:** asymptomatic → **leave the Foley catheter and drain in place**; symptomatic → **percutaneous drain**; before removing drain and catheter → **drain study and cystogram**; persistent leak → **nephrostomy tubes versus exploration and closure**.
- **Ureteric injury** may go unrecognised until after surgery, presenting as **acute kidney injury (AKI)**, features of extravasation (fever, abdominal pain, ileus, electrolyte disturbance) or flank pain.
  - **Excretory urography or retrograde imaging** confirms and characterises the injury, shows hydronephrosis and identifies a **urinoma**, which may need percutaneous drainage.
  - **Attempt retrograde stenting first; if it fails, place a nephrostomy and plan delayed repair**.
- **Other complications:** UTI, urinary fistula, bladder spasm, wound infection and bowel injury.

## Outcomes

- **Open surgery remains the gold standard**, but growing minimally invasive experience now allows treatment of even **complex, large or multiple diverticula**.
- The evidence consists mostly of **single-institution series**:

**MINIMALLY INVASIVE VERSUS OPEN DIVERTICULECTOMY — COMPARATIVE EVIDENCE**

| Comparison                                                                                                                                                                          | Finding                                                                                                                                                                                                    | Verdict                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Sequential transurethral resection of the prostate (TURP) then transperitoneal LBD, versus open diverticulectomy with transvesical prostatectomy (matched diverticula and adenomas) | <b>Longer operating time</b> laparoscopically, but <b>less blood loss, less pain, shorter stay</b> ; no complications in either arm; equivalent postoperative flow                                         | <b>Laparoscopy is a safe alternative</b> |
| Transperitoneal LBD without simultaneous outlet surgery; TURP performed shortly afterwards where BPH was the cause                                                                  | A single suture-line leak, managed conservatively; significant improvement in <b>International Prostate Symptom Score (IPSS)</b> and uroflowmetry; normal postvoid residual; no postoperative UTI          | Safe and effective                       |
| Sequential <b>holmium laser enucleation of the prostate (HoLEP)</b> and extraperitoneal LBD                                                                                         | Favourable outcomes                                                                                                                                                                                        | Feasible                                 |
| RBD series and a pooled review of RBD series                                                                                                                                        | <b>Short hospital stay</b> , low blood loss, few complications; catheter duration similar to open series; reductions in postvoid residual and in the American Urological Association Symptom Score (AUASS) | <b>Favourable, low-risk</b>              |

- **Bottom line:** LBD is a **safe and effective alternative to open diverticulectomy and may shorten hospital stay**; the aggregate RBD literature shows **favourable outcomes, a low complication risk, and operating times shorter than those of laparoscopy**.

**Concept** — Diverticulectomy is a **dissection problem, not a reconstruction problem**: find the sac (selective filling or transillumination), stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely — and fix the outlet first.

**KEY POINTS: BLADDER DIVERTICULECTOMY**

- Success depends on a **meticulous preoperative work-up** defining the number, size and location of diverticula — above all their **relationship to the ureter**, so that ureteral integrity is preserved.
- The **disease process that formed the diverticulum** must be managed before or at the time of repair.
- The **fibrous pseudocapsule** is the cleavage plane that frees the diverticulum from surrounding extravesical tissue.
- In minimally invasive surgery, **selective catheterisation of the diverticulum with a Councill-tip catheter over a guidewire** allows selective filling and intraoperative identification.
- Diverticula may be managed by **open, laparoscopic or robotic** techniques.

**URETERAL REIMPLANTATION****Description**

- In adults, reimplantation may be needed in both **acute and elective** settings.
- **Urgent indications:**
  - **distal ureteral trauma** — **open repair is recommended**;
  - **iatrogenic injury recognised during pelvic or retroperitoneal surgery** — open or minimally invasive reconstruction, depending on the approach of the index operation and the magnitude of the injury.
- **Elective indications (commonest):** **ureteral stricture disease**, and **unrecognised iatrogenic injury with delayed clinical presentation**.

- **Primary VUR is rarely the indication in adults**, in contrast to children.
- Reimplantation is also done **single-stage** within another planned operation — bladder diverticulectomy, partial cystectomy, or **distal ureterectomy for upper tract urothelial carcinoma**. The techniques below concern non-malignant indications.

## Evaluation and Surgical Indications

- Planning is paramount because **success depends on a tension-free anastomosis with the bladder**; the **length and location of the defect** must therefore be defined accurately.
- **Acute setting**: direct inspection usually defines the injury, so intraoperative imaging may be unnecessary.
  - Attend to the **health of the ureteral tissue and its periureteric blood supply**.
  - **Mechanism matters**: after **thermal or high-velocity projectile injury**, debridement of ureter that looks normal may be required.
  - For **incomplete transection or contusion**, an intraoperative **retrograde pyelogram** shows the presence, extent and site of extravasation; alternatively **methylene blue** is injected into the ureter proximally.
- **Elective setting**: use both **antegrade and retrograde imaging** to define the length and site of the diseased segment.
  - Antegrade imaging is by **CT urography**, or by **antegrade nephrostography** if a nephrostomy is present.
  - Note **multiple diseased segments** — they lengthen the overall defect even when intervening ureter looks normal.
- **Stricture causes**: an impacted stone, previous pelvic or abdominal **radiation**, chronic infection or inflammation.
- **Exclude malignancy before intervention**: urine cytology and, in **high-risk patients (smokers, older age)**, **ureteroscopy with biopsy**; the resected stricture can be sent for **intraoperative frozen section**.
- **Renal atrophy** on imaging, especially with chronic obstruction, calls for **nuclear renography** to document both obstruction and function. With severe obstruction a **nephrostomy** allows function to be reassessed after decompression.
  - **Do not reimplant if the ipsilateral kidney functions poorly**; perform **nephrectomy** instead if treatment is warranted.
- **Stent already in place** for contusion or partial transection: perform **retrograde pyelography or CT urography before removing the stent**; at retrograde study, exchange the stent for a guidewire to keep access across the bladder and ureter.
  - **Persistent extravasation despite an adequate period of stenting indicates persistent injury and is an indication for reimplantation**.
  - Define it by passing a ureteral catheter to the renal pelvis over the wire and injecting contrast during gradual withdrawal, opacifying the whole ureter.

**Concept** — *The whole work-up measures **one distance** — from healthy, well-vascularised ureter to bladder — and makes **two exclusions**: malignancy, and a kidney not worth saving.*

## Surgical Technique

- Because most adult cases are for injury, stricture, or reimplantation during partial cystectomy or diverticulectomy, **dissection of the intramural ureter is generally unnecessary**. The ureter is **divided proximal to the diseased portion and the distal stump tied**.
- Where ureteral length is inadequate, **advancement flaps from the bladder wall** achieve a tension-free anastomosis.

- **A nonrefluxing anastomosis is generally unnecessary in adults**, because VUR is rarely the indication. Studies show that a **refluxing anastomosis does not accelerate renal decline or increase stenosis** compared with a nonrefluxing one. Infection risk is less studied, but long-term adult data do not suggest an increase.

#### REFLUXING VERSUS NONREFLUXING REIMPLANTATION IN ADULTS

|                          | Nonrefluxing (tunnelled)                                    | Refluxing (direct)                                                |
|--------------------------|-------------------------------------------------------------|-------------------------------------------------------------------|
| Submucosal tunnel        | <b>Yes — 3–5 times ureteric diameter</b>                    | No                                                                |
| Ureteral length required | <b>Greater</b>                                              | <b>Less</b>                                                       |
| Bladder mobilisation     | Greater                                                     | Less — ureter goes to the accessible <b>posterior dome</b>        |
| Place in adults          | Needed chiefly for <b>high-grade reflux with renal risk</b> | <b>Generally sufficient</b> ; no excess renal decline or stenosis |

- **Cardinal principles, whatever the technique:**
  - **Handle the ureter delicately** to minimise devascularisation.
  - During extravesical mobilisation, **preserve the periureteric fibroadipose tissue and avoid skeletonisation**, which compromises blood supply.
  - **A tension-free ureterovesical anastomosis is paramount. Primary reimplantation is generally feasible for distal defects up to 4 cm**; for longer defects consider **psoas hitch and/or Boari flap**. **Renal mobilisation with downward nephropexy can add up to 5 cm**.
  - **Avoid excessive angulation** at the point of entry into the bladder, which causes postoperative obstruction.
  - **Insert the ureter along a path directed toward the bladder neck**.
- **Choice of approach: open surgery is preferred for trauma**, because it allows direct inspection of abdomen and pelvis for associated injuries. **Iatrogenic injury recognised intraoperatively** may be repaired by the **same approach as the index procedure**, depending on the surgeon's comfort and the set-up; in robotic or laparoscopic cases, add ports as needed.

#### Open Technique

##### *Ureteral mobilisation*

- Supine, ASIS over the table break, slight flexion; genitalia and abdomen prepared to just above the umbilicus; **16 Fr catheter** on the field.
- Lower midline incision to the symphysis → rectus fascia at the linea alba → recti split → transversalis fascia incised into the space of Retzius, **taking care not to enter the peritoneum**.
- The bladder is identified by palpating the **Foley balloon**, and the lateral gutters are developed bluntly with a sponge-stick, sweeping perivesical fat medially.
- **Finding the ureter:** exposing the great pelvic vessels helps. Trace the **external iliac artery proximally to the common iliac bifurcation**, where the ureter crosses from lateral to medial; alternatively find it **beneath the obliterated umbilical artery** near its origin from the proximal internal iliac artery.
- Dissect around the ureter with a right-angle clamp and pass a **vessel loop**; mobilise caudally and cephalad **without overdissection or devascularisation**.
- **For stricture or injury:** identify and resect the diseased segment.
  - **Clip the distal stump before transection**.

- A clip across the ureter **proximal** to the disease allows **passive ureteral distension**, which eases reimplantation.
- Place a **4-0 stay suture at the 6 o'clock position** on the ureteric end for handling.
- **In stricture disease, send a margin of the transected ureter for frozen section** to exclude malignancy.

#### Bladder mobilisation

- Extend the gutter dissection superiorly to displace the peritoneal reflection off the dome, sweeping the sponge-stick **cephalad and medially**; this may require dividing the **ipsilateral medial umbilical ligament and the urachus**.
- **A refluxing anastomosis needs less mobilisation** than a nonrefluxing one, because the ureter goes to the **posterior dome** — more accessible than the site a tunnelled anastomosis needs.
- **Test feasibility of primary ureteroneocystostomy** by laying the ureter against the **decompressed dome** and confirming no tension. If length is short, the ureter can be mobilised further cephalad — at the price of **devascularisation and later stricture**.
- Primary ureteroneocystostomy can then be performed **extravesically or intravesically, refluxing or nonrefluxing**:

#### THE FOUR OPEN REIMPLANTATION TECHNIQUES

| Technique                                                       | Access                     | Tunnel                                                                         | New hiatus                                                             | Signature step                                                                          |
|-----------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Nonrefluxing extravesical (modified Lich-Gregoir)</b>        | Outside the bladder        | <b>Detrusor trough</b> over intact urothelium, 3–5 × ureteric diameter         | Posterior wall, <b>superior and medial</b> to native orifice           | Serosal–detrusor flaps closed <b>over</b> the anastomosed ureter                        |
| <b>Nonrefluxing transvesical (modified Politano-Leadbetter)</b> | Anterior midline cystotomy | <b>Submucosal tunnel</b> cut from inside with Metzenbaum scissors, 3–5 × lumen | New posterior-wall entry <b>cephalad and medial</b> to native junction | Right-angle clamp through the posterior wall draws the ureter <b>through the tunnel</b> |
| <b>Refluxing extravesical</b>                                   | Outside the bladder        | None                                                                           | <b>Posterior dome, lateralised</b> to the side of pathology            | Direct spatulated anastomosis                                                           |
| <b>Refluxing transvesical</b>                                   | Anterior midline cystotomy | None                                                                           | Posterior dome, lateralised, made <b>over a Kelly clamp</b>            | Kelly clamp pulls the ureter in                                                         |

#### Nonrefluxing reimplantation — extravesical approach (modified Lich-Gregoir)

- **Partially distend the bladder** with sterile water — this aids both reimplantation and dissection of the bladder off its peritoneal attachments. Mobilise until **medial retraction of the bladder shows the native ureterovesical junction (UVJ)**.
- **Site:** posterior wall, **superior and medial to the native UVJ**; a more **lateral site risks kinking**, especially on bladder filling.
- Sequence:
  - **transmural cystotomy** at the neo-orifice, the mucosal defect **slightly wider than the ureteric lumen** →
  - **longitudinal incision through serosa and detrusor, sparing the urothelium, directed toward the bladder neck** →
  - detrusor dissected off the urothelium perpendicularly, creating a submucosal tunnel from outside; **tunnel length three to five times the ureteric diameter** →
  - any inadvertent urothelial breach closed with 4-0 polyglactin →

- distal ureter excised; the new end **spatulated anteriorly** with tenotomy scissors →
- apex anchored to the neo-orifice mucosa at **12 o'clock** (4-0 polyglactin), posterior ureter at **6 o'clock**, then interrupted sutures circumferentially →
- **before completion, a 6 Fr × 26 or 28 cm double-J stent is placed retrograde** across the anastomosis →
- the ureter is laid in the tunnel and the **serosal-detrusor flaps closed over it** with 2-0 polyglactin.
- **Suture technique:** bites through the ureter are **full-thickness (transmural)**; bites in the bladder take **mucosal edge plus some detrusor**, ensuring mucosal apposition; **knots lie outside** the anastomosis.
- **Finish:** large-bore Foley, fill to test; pelvic drain beside the anastomosis via a separate stab, secured with nonabsorbable suture, on self-suction; close fascia and skin.

#### *Nonrefluxing reimplantation — transvesical approach (modified Politano-Leadbetter)*

- Mobilise the bladder as for the extravesical approach; with the bladder partly distended, make a **midline anterior cystotomy**.
- Choose the neo-orifice on the posterior wall **superior and medial to the native orifice**; score the mucosa circumferentially to make a defect **slightly larger than the ureter (about 1.5 cm)**.
- Raise a **submucosal tunnel with Metzenbaum scissors** starting at the neo-orifice, **oriented longitudinally toward the bladder neck, three to five times the ureteric lumen** in length.
- Exchange the scissors for a **right-angle clamp**, beak directed posteriorly, and incise the posterior wall onto it — this is the **new entry point, cephalad and medial to the native UVJ**, which lets the ureter run almost longitudinally through the tunnel.
- Advance the clamp through the wall, grasp the ureteric stay sutures, and **draw the ureter through the tunnel and out of the neo-orifice**.
- Spatulate and anastomose to the neo-orifice with 4-0 polyglactin and stent as above; large-bore Foley; **two-layer 2-0 closure** of the cystotomy; fill to test; pelvic drain on self-suction; close.

#### *Refluxing reimplantation — extravesical approach*

- Grasp the dome with **Allis clamps**; make a **transmural cystotomy in the posterior dome lateralised to the side of pathology**, wide enough to take the ureter.
- Excise the distal ureter, **spatulate anteriorly** and anastomose as for the nonrefluxing extravesical technique.
- Large-bore Foley and pelvic drain; **irrigate to test** the anastomosis.

#### *Refluxing reimplantation — transvesical approach*

- With the bladder partly distended, make an anterior midline cystotomy. A **Kelly clamp** is passed through it and pressed against the **posterior dome, lateralised** to the side of pathology; a cystotomy is made over the clamp large enough for the ureter.
- The clamp is advanced through the defect, grasps the ureteric stay sutures and draws the ureter in as it is withdrawn — creating the **new UVJ and neo-orifice**.
- The ureter is spatulated and anastomosed to the neo-orifice mucosa; large-bore Foley; **two-layer 2-0 cystotomy closure**; fill to test; **closed-suction pelvic drain** through a separate stab; close.

#### **Laparoscopic and Robotic Approaches**

- Minimally invasive distal ureteral reconstruction is **widely used**; accumulated experience now **nearly replicates open technique**.
- A **transperitoneal extravesical approach** gives a wide working space, permits **both refluxing and nonrefluxing** anastomoses, and — because **psoas hitch and bladder advancement flaps are feasible** minimally invasively — allows experienced surgeons to manage **longer defects**.

- **Laparoscopic reimplantation uses three trocars** (left-sided example): a **10-mm umbilical camera port**, a **12-mm working port midway between umbilicus and symphysis**, and a **5-mm working port at the lateral rectus edge at umbilical level** on the side of the procedure.
- **Robotic** positioning and ports follow robotic diverticulectomy; **assistant ports go on the side contralateral to the ureteric pathology**, for more direct access.
- **Dissection sequence:**
  - identify the ureter **medial to the obliterated umbilical ligament** and incise the overlying peritoneum — or incise more proximally where it crosses the **common iliac bifurcation** →
  - mobilise caudally toward the UVJ, **preserving periureteric tissue** →
  - the **ipsilateral vas deferens (men) or broad ligament (women)** may be divided to expose the distal ureter →
  - an incision **lateral to the ipsilateral obliterated umbilical ligament** enters the space of Retzius and allows **medial rotation of the bladder**, which stays suspended because the **contralateral ligament is left intact** →
  - a **Hem-o-lok clip** closes the distal stump at the edge of the diseased segment (if possible); the segment is excised; **frozen-section margin in stricture disease** →
  - a clip on the proximal ureteric end allows passive dilatation; a **4-0 stay suture** allows atraumatic handling →
  - the bladder is **dropped** into the space of Retzius, and tension is assessed by bringing the ureter to the **ipsilateral dome of the non-distended bladder**.
- **Single-port robotic reimplantation** follows the same steps through one incision, which can give intraperitoneal, extraperitoneal or transvesical access; **intraperitoneal access is commonest**, because psoas hitch and extra bladder and ureteral mobilisation may be needed.

*Nonrefluxing reimplantation — extravesical (modified Lich-Gregoir)*

- Mark the tunnel with the bladder **filled with about 250–300 mL of saline**; orient it **toward the bladder neck**, at a site **superior and medial to the native UVJ** (lateral sites kink on filling).
- Keep a **5:1 tunnel-to-ureteric-diameter ratio**; incise the detrusor to make an adequate **trough without perforating the mucosa**; make a cystotomy at the distal end of the tunnel as the neo-orifice.
- Spatulate the ureter along its anterior surface; anchor the apex at 12 o'clock and the posterior ureter at 6 o'clock (4-0 polyglactin), then complete with interrupted sutures — **transmural through ureter, mucosa plus detrusor on the bladder, knots outside the lumen**.
- Before completion pass a **6 Fr × 26 or 28 cm double-pigtail stent** retrograde over a guidewire.
- Lay the ureter in the trough and **close the detrusor over it** with interrupted 2-0 absorbable sutures, completing the antireflux mechanism.
- Large-bore Foley, fill to test; drain through the lateral port on self-suction; release pneumoperitoneum; **close all 12-mm port fascia** with figure-of-eight 0-polyglactin; close skin.

*Refluxing reimplantation — extravesical*

- **Less ureteral length is needed** without a tunnel, so the ureter can be **mobilised less**.
- A **refluxing, non-transecting side-to-side reimplantation** has been described for distal strictures and may be advantageous over the standard refluxing technique in patients with **impaired vascularity**.
- A new hiatus is made at the **anterior dome, slightly lateralised** toward the pathology; the spatulated ureter is reimplanted with interrupted 4-0 absorbable sutures.
- Large-bore Foley, fill to test; drain via the lateral port; close 12-mm port fascia with figure-of-eight 0-polyglactin.

## Postoperative Care and Complications

- **Foley catheter for about 1 week; cystogram before removal.**
- **Pelvic drain** usually removed after **48 hours** once output falls; drain creatinine is optional and must be read in the context of output.
- **Ureteral stent for about 6 weeks**, removed cystoscopically.
- **Urinary extravasation** (open or minimally invasive) presents with fever and abdominal pain, particularly if the urinoma is infected; **intraperitoneal urinary ascites** causes ileus and electrolyte disturbance.
  - Causes: **a non-watertight anastomosis, or ischaemia** at the anastomosis.
  - **Excretory urography** defines site and degree and shows a urinoma, which may need **percutaneous drainage**.
  - Management: **keep the Foley and the stent** for maximal antegrade drainage; if the leak appears **after stent removal, replace the stent**; if refractory, consider a **percutaneous nephrostomy**.
- **Ureteral obstruction is a late complication, typically after stent removal** — found incidentally (hydronephrosis on ultrasound) or with flank pain or pyelonephritis.
  - Causes: **ureteral ischaemia from excessive skeletonisation, tension on the anastomosis, a narrow neo-orifice, inadequate anterior spatulation, or prior radiation**; less often, **a lateral reimplant site that kinks on filling**.
  - Define the stricture with **antegrade and retrograde imaging**. **Endoscopic management first, especially for short strictures**; **repeat surgery for high-grade obstruction**.
- **Other complications:** UTI, urinary fistula, bladder spasm, **stent colic**, wound infection, bowel injury.

## Outcomes

- **Open ureteroneocystostomy remains the gold standard for distal ureteral disease in adults**, because of its versatility: it can stay **entirely extraperitoneal**, reimplant **extravesically or transvesically**, construct a **refluxing or nonrefluxing** anastomosis, and readily incorporate **psoas hitch or Boari flap** for a tension-free repair.
- Large long-term series exist mainly for **open surgery**, showing favourable outcomes in the great majority — hydronephrosis resolved in most, although a minority still needed **long-term stenting**, and leak requiring prolonged drainage was uncommon.
- Minimally invasive experience continues to grow; small feasibility studies predominate, but a few larger intermediate-term studies support efficacy.

## MINIMALLY INVASIVE URETERAL REIMPLANTATION — COMPARATIVE EVIDENCE

| Comparison                                                        | Finding                                                                                                                               | Verdict                       |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Laparoscopic reimplantation for distal stricture (largest series) | No open conversions, modest blood loss, short stay, low-grade leaks managed conservatively; high success at 2 years                   | <b>Effective</b>              |
| Laparoscopic reimplantation after hysterectomy injury             | Durable ureteral patency                                                                                                              | Effective                     |
| Laparoscopic versus open (matched stricture series)               | Laparoscopy: <b>less blood loss and analgesia, earlier oral intake, shorter stay and convalescence</b> ; comparable or better success | <b>Laparoscopic advantage</b> |
| Laparoscopic versus open (retrospective)                          | Less blood loss, shorter stay; <b>equivalent complications and patency</b>                                                            | Laparoscopic advantage        |
| Robotic versus laparoscopic (benign stricture)                    | <b>Comparable</b> perioperative outcomes; occasional conversions in both; all successful                                              | <b>Equivalent</b>             |

| Comparison                                       | Finding                                                                                                                                                    | Verdict                               |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Robotic versus open (matched; and larger series) | Robotic: <b>much less blood loss, shorter stay, less opioid use</b> ; operating time similar or longer; <b>no difference in reoperation</b> for recurrence | <b>Robotic is a valid alternative</b> |
| Pooled minimally invasive versus open            | Lower blood loss, shorter stay, <b>fewer complications</b>                                                                                                 | Minimally invasive favoured           |
| Robotic versus laparoscopic (pooled)             | Robotic: shorter operating time and stay; <b>no difference in complications or success</b>                                                                 | Slight robotic edge                   |
| LESS and single-port robotic reimplantation      | Technically feasible; SP comparable early results to multiport robotics; LESS demanding, done by surgeons with broad LESS experience                       | <b>Larger series needed</b>           |

- **Why the robot appeals:** ergonomics, **three-dimensional magnification** and **articulating instruments** flatten the learning curve for **intracorporeal suturing**.

**Concept** — *Reimplantation fails for one reason in two forms — **ischaemia or tension** — and every technical rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) exists to prevent one or the other.*

#### KEY POINTS: URETERAL REIMPLANTATION

- **Avoid aggressive skeletonisation** during mobilisation — it causes ischaemia and stricture.
- A **tension-free anastomosis** is critical; tension risks anastomotic stricture.
- **Spatulate the ureter adequately** before anastomosis.
- In a nonrefluxing reimplant, the **submucosal tunnel should be three to five times the ureteric calibre**.
- Longer defects can be bridged by manoeuvres that allow **cephalad displacement of the bladder** — dividing the **contralateral medial umbilical ligament**, dissecting the **peritoneal reflection off the dome**, or ligating the **contralateral bladder pedicle**.

## PSOAS HITCH AND BOARI FLAP

### Description

- **Minimal distal ureteral loss** is managed by simple ureteroneocystostomy; **larger gaps** need more complex reconstruction.
- **Psoas hitch** (first described in **1960** and developed in **1969**): the bladder is drawn **cephalad** far enough for a tension-free anastomosis with healthy ureter.
- If the hitch alone is not enough, a **Boari flap** is added to bridge the gap — typically when disease extends **up to the mid-ureter**.
- **Boari flap history:** canine model **1894**, humans **1947**; it can bridge gaps **up to 15 cm**. **Laparoscopic Boari flap 2001, robotic 2008**; the robot eases the extensive intracorporeal suturing required.
- These techniques serve mostly the **distal and middle thirds**; in selected cases they reach the proximal ureter or renal pelvis, although **ureteroureterostomy, transureteroureterostomy, renal autotransplantation or ileal ureteric substitution** should then also be considered.

### Evaluation and Surgical Indications

- Indicated when ureteral loss **precludes simple ureteroureterostomy or ureteroneocystostomy**.

- Evaluation as for ureteroneocystostomy, including **antegrade and retrograde urography** to estimate the extent of disease and plan the approach.
- **Contraindication: a small-capacity, poorly compliant bladder.**
- **Intraoperative findings ultimately decide** whether a hitch, a flap, or both are required.

#### OPTIONS FOR BRIDGING A DISTAL URETERAL DEFECT

| Option                                   | Principle                                                         | Reach                                                        | Governing rule                                                                                                                     |
|------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Primary ureteroneocystostomy             | Direct reimplantation                                             | <b>Distal defects up to 4 cm</b>                             | Tension-free, directed toward bladder neck                                                                                         |
| <b>Psoas hitch</b>                       | Bladder dome fixed to the <b>psoas minor tendon</b>               | Gains <b>up to 5 cm</b>                                      | Sutures <b>superficial and parallel to the fibres</b> ; reimplant <b>before tying</b>                                              |
| <b>Boari flap</b>                        | Tubularised anterior bladder-wall flap                            | Gaps <b>up to 15 cm</b> (to mid-ureter, occasionally higher) | <b>Base <math>\geq 4</math> cm, apex <math>\geq 3</math> cm, length:base <math>\leq 3:1</math></b> , measured on the empty bladder |
| <b>Bladder advancement flap</b>          | Transverse incision one-third from dome to neck; bladder advanced | <b>Comparable to Boari</b>                                   | Simpler; may replace the Boari flap                                                                                                |
| Renal mobilisation / downward nephropexy | Kidney brought down                                               | <b>Up to 5 cm</b>                                            | Adjunct when other measures fall short                                                                                             |

- **What the table tests:** the ladder is climbed in order — reimplant, then hitch, then flap — and the hitch is combined with a flap so that the flap can be shorter and bladder capacity preserved.

## Surgical Technique

### Open Technique

#### *Psoas hitch*

- Mobilise the bladder on the **side of the ureteral pathology** as for reimplantation. Draw the ipsilateral dome **cephalad over the ipsilateral psoas above the common iliac bifurcation**; lay the mobilised proximal ureter on the **superolateral dome** to confirm a tension-free anastomosis is possible.
- **Extra bladder mobility:** divide the **contralateral round ligament (women) or vas deferens (men)**; free the contralateral perivesical attachments, which may involve **ligating the contralateral superior vesical artery** (less often the middle vesical arteries) and opening the contralateral endopelvic fascia.
- Stay sutures on the anterior dome → **oblique cystotomy across the anterior dome at its widest diameter**, spanning **about half the anterior dome** (a shorter incision limits cephalad displacement) → the surgeon's finger inside the bladder pushes the ipsilateral posterolateral dome onto the psoas above the iliac bifurcation → if more length is needed, **horizontal incisions** off the oblique cystotomy release the dome further.
- **Fixation:** four to six 2-0 polyglactin sutures through the **psoas minor tendon** and detrusor, placed **superficially and parallel to the muscle fibres** to avoid the **genitofemoral nerve (on the surface of psoas)** and the **femoral nerve (deep in the muscle belly)**.
- **Before tying the hitch sutures, choose the reimplant site and deliver the ureter into the bladder.** The ideal site is the **superolateral aspect of the displaced dome** — the ureter follows its native path onto a **relatively immobile** part of the bladder, reducing kinking.
- **Nonrefluxing:** submucosal tunnel by an extravesical or intravesical approach, as described earlier. **Short ureteral segments may require a refluxing anastomosis.**

- Extravesical **4-0 polyglactin** sutures loosely anchor the **ureteric adventitia to the detrusor** at its entry; a stent is placed; the cystotomy is closed in **two layers of 2-0 polyglactin** (mucosa first, then detrusor).

### *Boari flap*

- **Outline an oblique flap** across the anterior bladder wall with its **base on the side of the pathology**.
- **Dimensions:**
  - **base at least 4 cm wide**, wider still for longer flaps, to keep the blood supply;
  - **apex at least 3 cm wide**;
  - **length approximately equal to the ureteral defect**, plus usually **3 cm more if a tunnel is planned** for a nonrefluxing anastomosis;
  - **length-to-base ratio must not exceed 3:1** — beyond it the flap becomes ischaemic;
  - **measure with the bladder non-distended**.
- Incise the outline with electrocautery and **raise the flap up to the ureter**; **anchor its base to the psoas tendon** as for a hitch. **Combining the flap with a psoas hitch shortens the flap needed and preserves bladder capacity**.
- **Nonrefluxing anastomosis:** allow **about 3 cm of overlap** between flap and ureter; raise a **submucosal tunnel about 3 cm long** (extra- or intravesically); bring the ureter **through the posterior wall of the flap, off tension**; after anastomosis, **tubularise the flap** and close in two layers of 2-0 polyglactin; anchor the ureteric adventitia to the flap's posterior wall with 4-0 polyglactin.
- **Refluxing anastomosis:** sew the spatulated ureter directly to the mucosal edges of the tubularised flap with **4-0 poliglecaprone**; to avoid stenosis, the **flap apex should be about three times the ureteric diameter**.
- Close the cystotomy in two layers of 2-0 polyglactin.
- If the flap still falls short: further ureteral dissection (avoiding overdissection), and **mobilising the kidney downward — up to 5 cm more**.

### **Laparoscopic and Robotic Approaches**

#### *Psoas hitch*

- Laparoscopic ports as for laparoscopic reimplantation; robotic ports as for diverticulectomy; **transperitoneal approach preferred**.
- Sequence:
  - incise the ipsilateral **white line of Toldt**, reflect the colon medially, and find the ureter where it crosses the iliac vessels →
  - mobilise distally and excise the diseased segment, ensuring the **distal margin is healthy and well vascularised**; **frozen section in stricture disease**; clip the transected end for passive dilatation →
  - fill the bladder with saline; mobilise it by incising peritoneum **lateral to both obliterated umbilical ligaments** and dividing the urachus; the bladder is "dropped" into the space of Retzius →
  - suture the **posterior bladder wall to the psoas minor tendon** (found **superior and lateral to the ipsilateral common iliac vessels**) with 2-0 polyglactin, bites **superficial and parallel to the fibres** in the tendon and **through detrusor** in the bladder →
  - **contralateral pedicle division is usually unnecessary**; if more mobility is needed, the contralateral superior and middle vesical arteries can be ligated and divided →
  - reimplant the ureter on the **superolateral displaced dome, refluxing and extravesically**, with a retrograde stent placed before completion →
  - fill the bladder to find leaks, reinforce as needed, and place a pelvic drain through a lateral trocar.

*Boari flap*

- **First laparoscopic Boari flap 2001** (in three patients with distal ureteral obstruction); **robotic-assisted Boari reported later (2008)**. Ports as above.
- **Ligation of the contralateral superior vesical artery plus psoas hitch often gives enough mobility** for ureteroneocystostomy; if not, a Boari flap or advancement flap is made.
- The anterior flap is cut with electrosurgical scissors or a 10-mm LigaSure, **beginning about 2 cm from the bladder neck and extending to the ipsilateral dome; apex about 3 cm, base about 4 cm**, checking that the base is wide enough for vascularity. The bladder can be distended with methylene blue to show the outline.
- Raise the flap cephalad; **suture its base to the psoas tendon (or muscle)**; spatulate the ureter and anastomose it to the flap apex with **interrupted 4-0 absorbable sutures** (a refluxing anastomosis).
- Place a **7 Fr double-J stent over a guidewire**; close the flap in two running layers of 2-0 polyglactin (or with the Endo Stitch device), tubularising it; fill to test; closed-suction drain through a lateral trocar.

*Bladder advancement flaps*

- The **laparoscopic bladder advancement flap (2005)** was introduced as a **simplified alternative to the Boari flap**.
- The bladder is opened by a **transverse incision on the anterior surface one-third of the distance from the dome to the bladder neck**; as the incision widens, the flap is advanced to the ureter, and the spatulated ureter is anastomosed to it as for a Boari flap.
- Because of its **simplicity, versatility, and reach comparable to the Boari flap**, an advancement flap may be used **instead of a Boari flap** for longer defects not amenable to primary reimplantation or psoas hitch — a large ("mega") advancement flap can reach even a defect extending to the proximal ureter.

**Postoperative Care and Complications**

- Management and complications **follow those of ureteroneocystostomy**.
- **Additional complications:**
  - **Neurapraxia of the genitofemoral or femoral nerve** from hitch sutures — prevented by **superficial sutures in the tendon, in the direction of the fibres**.
  - **Iliac vessel injury** as the bladder is drawn over the common iliac bifurcation.
  - **Ischaemic anastomotic stricture** of a Boari flap with a **narrow base relative to its length** — prevented by respecting the **3:1 ratio**.
  - **Reduced bladder capacity** after a Boari flap, with irritative symptoms: **urgency, frequency and incontinence**.

**Outcomes**

- **Open psoas hitch** has been used since the mid-20th century to gain **up to 5 cm** of length. Long-term series, mostly of **operative ureteral injury**, show **high success** (no further intervention, symptom relief, stable renal function), with **no difference between immediate and delayed repair**.
- The **Boari flap bridges defects approaching 15 cm**, sparing the patient more complex, higher-risk procedures such as **ileal interposition or autotransplantation**; series of mostly gynaecological injuries show resolution of obstruction in all, with high success and low morbidity in contemporary series.
- **Laparoscopic psoas hitch** reproduces open results — including for ureteral **endometriosis** and gynaecological injury — with no obstruction or reflux at follow-up; **proper intracorporeal suturing technique is stressed as critical**. Contemporary laparoscopic series now report hitch cases together with primary reimplants, showing it has entered the routine minimally invasive repertoire.

- **Laparoscopic Boari flap:** first series showed feasibility; larger series confirmed **high success**, with complications including anastomotic leak, pulmonary embolism, small-bowel obstruction and *Clostridioides difficile* colitis; a multi-institutional series showed **no intraoperative complications or conversions and very high success**.
- **Robotic psoas hitch and Boari flap:** high efficacy with few or no complications; occasional **open conversion for dense adhesions**; most patients free of obstruction and symptoms; a multi-institutional robotic Boari series (including single-port cases) showed high success.
- **Bottom line:** laparoscopic and robotic series are small, but **in experienced hands minimally invasive results are excellent**.

**Concept** — *In distal ureteral reconstruction the **bladder is the donor tissue**: hitch it first, flap it second — and every flap rule (base width, the 3:1 ratio, measuring the bladder empty) exists to protect the flap's blood supply and the bladder's capacity.*

#### KEY POINTS: PSOAS HITCH AND BOARI FLAP

- Hitch sutures fixing the bladder to the **psoas minor tendon** are placed **superficially and parallel to the muscle fibres** to avoid entrapping the **femoral and genitofemoral nerves**.
- **Reimplant the ureter before tying** the psoas anchoring sutures.
- The **Boari flap base must be at least 4 cm**, wider for longer flaps; **length-to-base ratio not above 3:1**.
- A nonrefluxing anastomosis into a Boari flap needs **at least 3 cm of overlap** between flap and ureter.
- Owing to their simplicity and versatility, **advancement flaps may replace Boari flaps** for longer defects.

## ENTEROCYSTOPLASTY

### Description

- In adults, bladder augmentation is most often indicated for **neurogenic bladder (congenital or acquired) after more conservative attempts to improve compliance have failed**. Less often it follows **partial cystectomy** that has substantially reduced capacity.
- The **intrinsic distensibility of bowel** lowers intravesical pressure and restores reservoir function — **but only if the segment is adequately detubularised**, which abolishes the coordinated peristaltic contractions that would otherwise defeat the purpose.
- Traditionally open, the operation has gradually moved toward minimally invasive techniques; **ileocystoplasty has become the most widely used**. Laparoscopic, robotic and LESS augmentation are reproducible by **purely intracorporeal or partly extracorporeal** approaches.

### Evaluation and Surgical Indications

- As non-invasive treatment has improved, augmentation for neurogenic bladder has **declined**. The mainstay is **clean intermittent catheterisation (CIC)**, **anticholinergics**, **intravesical botulinum toxin injection** and **sacral neuromodulation**.
- **Augment when noncompliance persists despite these measures**, particularly with **upper tract deterioration, hydronephrosis, or incontinence severe enough to cause skin ulceration**.
- **Urodynamics:** the **detrusor leak point pressure** is a strong indicator of the bladder's elastic properties and of the **risk of renal deterioration**. **Video-urodynamics is preferred** — it shows bladder contour, associated anomalies (for example diverticula) and VUR at the same time.
- **Ultrasonography routinely:** **hydronephrosis or parenchymal atrophy indicates poor wall compliance**.

- **After partial cystectomy**, frequency, urgency, nocturia or incontinence may signal a significant loss of capacity. **Conservative measures first**, as for neurogenic dysfunction; **augment only for refractory symptoms**. Urodynamics characterises function and response; in **men**, **cystoscopy excludes concomitant outlet obstruction** from BPH or urethral stricture.
- **Metabolic screening**: bowel in the urinary tract can cause metabolic derangement, so obtain a **full serum chemistry**.
  - With ileum, screen carefully for **vitamin B<sub>12</sub> and fat-soluble vitamin deficiency**.
  - Urine–intestinal mucosa interaction disturbs pH, a special concern in **renal failure** where pH regulation is impaired. **Because ileum is used most, the principal concern is chronic metabolic acidosis and the resulting bone demineralisation.**
- **Bowel history**:
  - **Crohn disease or previous intestinal surgery** risks **short gut syndrome** after further resection.
  - **Prior radiation** may prevent isolation of a segment with a durable blood supply.
- **Catheterisation remains necessary in most augmented patients**, so **manual dexterity and anatomical ability to catheterise** must be confirmed beforehand, especially in neurogenic dysfunction.
- **Concomitant procedures**:
  - **Stress urinary incontinence (SUI)**: raise outlet resistance at the same sitting — **artificial urinary sphincter, urethral sling or bladder neck closure**.
  - **Continent catheterisable stoma**: bladder neck closure may be planned, particularly with incontinence.
  - **Nonrefluxing reimplantation for high-grade reflux** when there is renal dysfunction and concern that intravesical pressure may stay high despite augmentation.

#### BOWEL SEGMENTS FOR AUGMENTATION

| Segment            | Why chosen                                                                                         | Special point                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Ileum              | <b>Commonest</b> — abundant, easily mobilised (intraperitoneal, long mesentery), rich blood supply | Take <b>~20 cm proximal to the ileocaecal valve</b> ; risk of <b>metabolic acidosis, B<sub>12</sub> and fat-soluble vitamin deficiency</b> |
| Ileocaecal segment | When a <b>continent catheterisable channel</b> is wanted, especially after bladder neck closure    | —                                                                                                                                          |
| Sigmoid colon      | Attractive in neurogenic dysfunction — <b>redundant mesentery</b> eases mobilisation               | <b>Colonoscopy beforehand</b> to exclude malignancy                                                                                        |
| Stomach            | Possible                                                                                           | Incision carried higher; preparation extended                                                                                              |

- Nearly any part of the gastrointestinal tract can be used; the principles shown for ileum apply to all segments.

## Surgical Technique

### Open Technique

- Supine, table flexed at the ASIS; prepared from thigh to just above the umbilicus (higher if stomach is used).
- Midline incision from umbilicus to pubic symphysis → rectus fascia → recti separated at the linea alba → transversalis fascia → **peritoneum opened sharply** along the incision.
- Space of Retzius developed bluntly; **extensive lateral dissection is unnecessary** because the bowel patch goes onto the most cephalad part of the bladder.

- **Harvest sequence:**
  - isolate **25 cm of ileum about 20 cm proximal to the ileocaecal valve**, sparing the terminal ileum to reduce malabsorption →
  - bring it to the pelvis to confirm a **tension-free reach to the dome** →
  - **transilluminate the mesentery** to confirm an adequate arterial arcade reaching the ends of the segment →
  - mark the boundaries with loose 2-0 silk stay sutures on the antimesenteric border; make small mesenteric windows →
  - divide the bowel with an **Endo GIA stapler**; divide the mesentery toward its root with a 10-mm LigaSure or stapler, **avoiding vessels and keeping the mesenteric base broad to prevent torsion**.
- **Restore bowel continuity** with a **side-to-side stapled anastomosis**; close the elbow with a **TA stapler** after apposing the edges with Allis clamps; **close the mesenteric defect** with longitudinal 2-0 polyglactin sutures to prevent **internal herniation**. The anastomosed bowel lies above the isolated segment.
- **Prepare the patch:** excise the staple lines, **irrigate copiously, detubularise along the antimesenteric border** with electrocautery, fold into a **U**, and join its sides with running full-thickness 2-0 polyglactin.
- **Bivalve the bladder** by a longitudinal incision along the dome from anterior to posterior; the opening must be **wide enough for a wide-mouthed anastomosis**, which lets the patch function as reservoir.
- **Anastomose the U-patch** circumferentially to the bladder edges with **transmural sutures** through bowel and bladder.
- **Before completion:** a **suprapubic tube** through a separate cystotomy; if the bladder neck is open, a **large-bore Foley** under vision.
- **Test** by irrigation through the Foley and/or suprapubic tube; reinforce leaks with figure-of-eight 2-0 polyglactin; retropubic pelvic drain; close fascia and skin.

### Laparoscopic and Robotic Techniques

- Cystoscopy with placement of **bilateral open-ended or single-J ureteral catheters**, secured to a urethral catheter; pneumoperitoneum.
- **Three to five trocars** according to experience — for example an **umbilical port, one at each lateral rectus border at umbilical level, and one in the iliac fossa at the level of the ASIS** (four-port transperitoneal technique).
- Mobilise the bladder fully — incise peritoneum lateral to both obliterated umbilical ligaments, divide the urachus, **drop the bladder** into the space of Retzius.
- Choose a **15–20 cm segment** with an adequate vascular pedicle that **will reach the bladder neck**. Bowel work is done **purely laparoscopically or extracorporeally through an extended umbilical incision**.
- Harvest with the Endo GIA stapler; divide mesentery with further reloads or a 10-mm LigaSure; restore continuity by **side-to-side stapled anastomosis**, completed with a **TA stapler (extracorporeal)** or **Lembert sutures (intracorporeal)**; **close the mesenteric window ("trap")** to prevent internal herniation.
- Irrigate, excise the staple lines, open the segment along the antimesenteric border, and form the **U** with running 2-0 polyglactin.
- Fill the bladder and **bivalve it in the midsagittal plane**; bring the segment down **without tension or torsion** of its pedicle.
- Orient the segment with the **apex of the U anteriorly at the bladder neck**; sew it to the bladder with **running 2-0 polyglactin, beginning posteriorly and ending anteriorly** (continuous full-thickness freehand suturing).
- Confirm integrity by irrigation via the Foley; place a pelvic drain through a port.

- **Related procedures:** laparoscopic and robotic **ileocaecocystoplasty with a catheterisable stoma**; laparoscopic and robotic **appendicovesicostomy** as an alternative for selected patients; **LESS enterocystoplasty** has been reported, but larger series are needed.

## Postoperative Care and Complications

- **Foley catheter about 1 week; cystogram before removal.** Pelvic drain removed at **about 48 hours** once output falls; drain creatinine optional, read in context of output.
- **Complications shared with diverticulectomy:**
  - **Urinary extravasation** across the suture lines from **inadequate suturing or ischaemia of the intestinal flap** — hence the need to **preserve a robust vascular arcade** at harvest. Presentation and management are as for diverticulectomy.
  - UTI, bowel injury and urinary fistula.
  - **Intra-abdominal infection**, particularly from irrigation and detubularisation of the bowel if proper **isolation precautions** are not taken.
- **Complications unique to enterocystoplasty: perforation, stones, metabolic derangement and neoplasia.**
  - **Perforation: regular intermittent catheterisation is imperative — failure to catheterise risks perforation.** Suspect it with **abdominal pain, fever or acute distension**, and obtain a **CT cystogram** with cross-sectional imaging; keep a **high index of suspicion in neurogenic patients, whose sensory deficits may mask it. Perforation is a surgical emergency requiring exploration.**
  - **Dexterity decides the reconstruction:** a patient whose habitus prevents catheterisation at the bladder neck may be better served by a **continent ileocaecocystoplasty**; a patient unable to catheterise, or poorly compliant, can have an **incontinent ileovesicostomy**.
  - **Stones** are multifactorial: **mucus** from the bowel, heavier **bacterial colonisation** and infection, and **stasis**. **Irrigation** is thought to reduce retained mucus, though its true benefit is debated; treat symptomatic infections with antibiotics.
  - **Neoplasia:** increased cancer risk has been reported in augmented bladders — **both adenocarcinoma and urothelial carcinoma. Surveillance endoscopy is recommended from 10 years after surgery.**
  - **Routine blood tests** screen for metabolic derangement and vitamin deficiency.

## Outcomes

- **Open augmentation is widely used for neurogenic dysfunction;** because minimally invasive techniques are recent, **long-term data come mainly from open surgery.**
- **Open series — bottom lines:**
  - the earliest ileal and caecal series showed **high continence** at medium-term follow-up;
  - long-term (over a decade) follow-up after ileocystoplasty for spinal cord injury showed **greater capacity, lower pressure, very high satisfaction and no change in bowel or sexual function — but complications at some point were common, mostly bladder stones;**
  - another long-term cohort showed **near-universal resolution of incontinence**, larger capacity, a substantial fall in maximum detrusor pressure and near-universal satisfaction — but **about half needed further procedures.**
- **Verdict:** open augmentation is **effective** for neurogenic patients with refractory symptoms (incontinence, overactivity) or at risk of upper tract damage from high pressure; long-term complications are relatively frequent but **mostly low-morbidity and manageable by secondary intervention.**

**MINIMALLY INVASIVE AUGMENTATION — MILESTONES**

| Milestone                                                                                       | Bottom line                                                                                                      |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>First pure laparoscopic gastrocystoplasty (1995)</b>                                         | Feasible, but a very long operation and prolonged stay                                                           |
| Laparoscopic-assisted augmentation                                                              | Laparoscopy limited largely to bowel mobilisation                                                                |
| <b>First laparoscopic ileocystoplasty, sigmoidocystoplasty and caecocystoplasty (2000)</b>      | Bowel work extracorporeal, four ports; long operations; a rectus sheath haematoma the only complication          |
| <b>First purely laparoscopic ileocystoplasty (2002)</b>                                         | Long operation; prolonged ileus                                                                                  |
| Laparoscopic ileocystoplasty with an antegrade continence enema (ACE) procedure                 | Continent between catheterisations; near-perfect faecal continence; one anastomotic leak and one stomal stenosis |
| <b>Robotic intracorporeal ileocystoplasty, with or without Mitrofanoff appendicovesicostomy</b> | Technique essentially laparoscopic, but eases intracorporeal suturing; continent with normal upper tracts        |
| Larger robotic intracorporeal series                                                            | Low blood loss, occasional conversion; <b>increased capacity and reduced detrusor pressure</b>                   |

- **Bottom line:** data are less robust, but small series show feasibility and efficacy at intermediate follow-up, and **outcomes are improving with experience.**

**Concept** — Augmentation *exchanges a pressure problem for a lifelong bowel-in-bladder problem: detubularisation and a wide-mouthed anastomosis win the pressure battle, while catheterisation, irrigation, blood tests and endoscopic surveillance manage the price.*

**KEY POINTS: ENTEROCYSTOPLASTY**

- **Ileum** is used most often, owing to its abundance, ease of mobilisation and robust blood supply.
- Isolate ileum **about 20 cm proximal to the ileocaecal valve**, sparing the terminal ileum to avoid **bile salt and vitamin B<sub>12</sub> malabsorption**.
- **Detubularise** the segment to prevent coordinated peristalsis, which would otherwise raise intravesical pressure.
- Keep a **broad mesenteric base** to avoid torsion and preserve blood supply.
- **Bivalve the bladder generously** along its dome for a wide-mouthed anastomosis; a **narrow mouth creates a pseudodiverticulum** that empties poorly and causes complications.
- Counsel patients that **regular intermittent catheterisation is essential** — failure risks perforation.

**CONTINENT CATHETERISABLE CHANNEL****Description**

- A continent catheterisable channel made from a harvested bowel segment can be used in **children and adults** to improve bladder dynamics and **protect the upper tracts**, generally in **neuropathic bladder with adequate capacity that needs chronic catheterisation**.
- It also provides **alternative bladder access**, for example in **refractory urethral stricture disease**.
- Channels can be **combined with augmentation** when compliance or capacity is poor.
- **Channel types: appendicovesicostomy (Mitrofanoff) and retubularised ileum (Yang-Monti)**. Vas deferens, colon and stomach have been used, but are **technically challenging or associated with poorer outcomes**.

## Evaluation and Surgical Indications

- **Patient selection is paramount.** Take a thorough medical and surgical history: **inflammatory bowel disease** and **prior radiation**; previous **small-bowel resection or appendicectomy** may limit available bowel.
- **Physical examination** shows whether the patient has the **dexterity to catheterise** and whether **body habitus** limits options; understanding the neurological condition also shapes the recommendation.
- **Urinalysis, serum creatinine and upper tract imaging** establish baseline renal function and drainage.
- **Video-urodynamics** may be needed to confirm **adequate capacity and safe storage pressure** and to detect anatomical abnormalities or VUR.
- Candidates have usually **failed more conservative measures, including clean intermittent urethral catheterisation.**

## Surgical Techniques

- A channel needs **enough length** to reach from bladder to stoma site while **keeping its blood supply and avoiding angulation.** Common stoma sites: the **right lower quadrant or the umbilicus.**
- **Continence depends on submucosal tunnelling of the channel through a sufficient length of detrusor.**
- Every channel operation needs a **fallback plan** — the appendix may be absent or too short, or the ileal segment may not reach the skin.

### Appendicovesicostomy

- First described with a submucosal detrusor tunnel by **Mitrofanoff in 1980.**
- The appendix is **often ideal**: a natural hollow tube of the right calibre for catheterisation, whose use in the urinary tract **causes no metabolic derangement.**
- Access by lower midline or **Pfannenstiel** incision. The appendix is checked for **adequate length** to span bladder to stoma; if more is needed, the **caecum and ascending colon** are mobilised.
- The appendix is detached proximally **with a small cuff of caecum**, adding length and a wider stoma; the caecal opening is closed.
- The distal end is opened and **spatulated** for anastomosis to the bladder; **submucosal tunnelling is essential for continence** and can be done at various sites on the bladder, intravesically or extravesically.

### Reconfigured ileum (Yang-Monti channel)

- Used when the appendix is **absent, too short, or needed for another purpose** (such as a continence enema).
- A **2–4 cm segment of terminal ileum** is isolated and **detubularised on its antimesenteric border**; the ileum is re-anastomosed.
- The segment is **retubularised transversely over a catheter**, producing a **longer, narrower tube.**
- For more length a **second reconfigured segment** can be joined — but this is associated with **more channel diverticula and difficult catheterisation.** A **spiral Monti** also adds length but has a **higher revision rate** than the standard channel.

### MITROFANOFF VERSUS YANG-MONTI CHANNEL

|         | Appendicovesicostomy (Mitrofanoff)                         | Reconfigured ileum (Yang-Monti)                                            |
|---------|------------------------------------------------------------|----------------------------------------------------------------------------|
| Status  | <b>First choice when available and long enough</b>         | Used when the appendix is absent, short or reserved                        |
| Conduit | Natural tube of ideal calibre; <b>no metabolic effects</b> | <b>2–4 cm ileum</b> , detubularised then <b>retubularised transversely</b> |

|                        | Appendicovesicostomy (Mitrofanoff)               | Reconfigured ileum (Yang-Monti)                                                         |
|------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------|
| Adding length          | Caecal cuff; mobilise caecum and ascending colon | Second segment ( <b>channel diverticula</b> ) or spiral Monti ( <b>more revisions</b> ) |
| Minimally invasive use | Reported widely, mostly in children              | <b>Rarely done</b> — technically demanding                                              |
| Continence mechanism   | <b>Submucosal detrusor tunnel</b>                | <b>Submucosal detrusor tunnel</b>                                                       |

### Laparoscopic and robotic approaches

- **Yang-Monti channels are rarely done minimally invasively** because of technical difficulty, though laparoscopic and robotic versions exist. **Appendicovesicostomy** has been reported most extensively, mainly in children.
- Sequence:
  - lithotomy with **10–15 degrees of Trendelenburg**, allowing bladder access and keeping bowel out of the field →
  - intraperitoneal access and insufflation near the umbilicus; further ports under vision →
  - identify the appendix, place a stay suture for handling →
  - **detach the appendix from the caecum** and close the caecal defect →
  - spatulate the distal appendix; create a **submucosal detrusor tunnel of about 4 cm** at the dome or anterior wall →
  - pass the appendix through a **button-hole in the mucosa** and suture with **4-0 polydioxanone (PDS)** →
  - leave a feeding tube or catheter in the channel for recovery; **mature the stoma** in the right lower quadrant or umbilicus; closed-suction pelvic drain.
- **Evidence:** a multi-institutional paediatric robotic series showed **high continence with a modest revision rate**. Compared with open surgery, the robotic approach gave a **shorter stay and faster return to diet but longer operating time**, with **no difference in stenosis, complications or revision**. Single-port robotic **appendicovesicostomy** has been shown feasible in case reports.

### Postoperative Care and Complications

- Reporting is sparse; patients are generally **admitted**. Pelvic drains are monitored and removed before discharge; drain creatinine can be checked for high output.
- The bladder is drained continuously by **Foley or suprapubic catheter for about 2–4 weeks**; then the **stoma catheter is removed and intermittent catheterisation through the channel begins**.
- **Continence rates are high — generally above 95% — regardless of channel type.**
- **The commonest complication is stomal stenosis**, in both appendicovesicostomy and ileal channels.

**Concept** — *The continence of any channel lives in its **submucosal tunnel**, not in the conduit: use the appendix when it is present and long enough, and reconfigure ileum only when it is not.*

#### KEY POINTS: CONTINENT CATHETERISABLE CHANNEL

- When available and long enough, **use the appendix**.
- The two commonest channels — **appendicovesicostomy (Mitrofanoff)** and **reconfigured ileum (Yang-Monti)** — both depend on **submucosal tunnelling through the detrusor** for continence.

## PARTIAL CYSTECTOMY AND URACHAL SURGERY

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### Description

- Partial cystectomy is used mostly in selected patients with **muscle-invasive urothelial carcinoma or urachal adenocarcinoma** suitable for bladder preservation; it also has several **benign indications**.
- Most benign bladder tumours are managed **conservatively (removing the offending stimulus) or by transurethral resection (TUR)**. Tumours that **grow large, infiltrate locally, or cause refractory local symptoms** may need resection — most commonly **leiomyoma and paraganglioma**.
- Other benign indications: **bladder endometriosis, urachal anomalies, and colovesical or vesicovaginal fistulae** that have failed conservative management.

### Evaluation and Surgical Indications

- Selection criteria are **less stringent than for malignancy**: tumour multiplicity, premalignant lesions such as **carcinoma in situ (CIS)**, and recurrence history matter less.
- The **primary question is whether resection will reduce capacity enough to harm quality of life** — so avoid partial cystectomy in patients with a **small-capacity bladder or severe dysfunctional voiding** at baseline.
- **Malignancy must be excluded**, because the operation differs (for cancer: **wide margins and lymph node dissection**). A **cancer-type work-up** is therefore usual: **cross-sectional excretory-phase imaging, cytology, cystoscopy and, in most cases, biopsy** for histological diagnosis.
- **Tumour location** affects feasibility — lesions near the **bladder base** are harder; a lesion near the **ureteric orifice** may require **simultaneous reimplantation**.

#### *Leiomyoma*

- Rare **smooth-muscle tumour**, reported mainly in **women of childbearing age**; histologically like the uterine **fibroid**. Aetiology uncertain — **hormonal imbalance or chronic detrusor irritation** is suspected.
- Can grow large, causing **outlet or ureteral obstruction**, irritative voiding symptoms, pelvic pain and **haematuria**.
- Growth may be predominantly **endovesical, intramural or extravesical** (a well-circumscribed lesion within the detrusor on CT); **transvaginal ultrasound**, and less often MRI, characterises it.
- **Small endovesical lesions → TUR; large or intramural lesions → wide local resection** to minimise regrowth.

#### *Paraganglioma*

- An **extra-adrenal tumour of sympathetic chromaffin cells**, commonest along the sympathetic chain; benign bladder tumours occur.
- **Most are functional**, releasing catecholamines — episodic **hypertension, tachycardia, palpitations, sweating or tremor**. **Unique to bladder paraganglioma: hypotension and syncope with micturition or sexual activity**.
- Despite this, they often present **non-specifically** — **painless haematuria**, or bladder wall thickening found incidentally on cross-sectional imaging.
- **Instrumentation or TUR without preoperative  $\alpha$ -adrenergic blockade** can provoke a **dangerous intraoperative catecholamine surge**, especially in asymptomatic patients.
- **If suspected: plasma metanephrines and functional imaging** — a positive **metaiodobenzylguanidine (MIBG)** scan confirms a functional mass.
- Cystoscopically: **yellow areas of raised mucosa**, because the tumour lies within the detrusor.

- **Partial cystectomy is preferred to TUR even for benign lesions, because of a high local recurrence rate after incomplete excision.**

### Endometriosis

- Bladder endometriosis may be **primary** or **iatrogenic** (dissemination at pelvic surgery).
- Classic symptoms: **cyclical haematuria or irritative voiding symptoms coinciding with menses.**
- Involvement is **generally extravesical**, so **cystoscopy is of limited value** (it shows only mucosal involvement — a bluish submucosal lesion with neovascularisation). **Transabdominal ultrasound, often with transvaginal ultrasound**, gauges size and depth; **MRI** delineates extent (high signal on T1-weighted images).
- **Partial cystectomy is the primary treatment**, again because of the extravesical distribution.

### Urachal abnormalities

- The **urachus**, the embryological remnant of the allantois, provides ligamentous support from dome to umbilicus. Urachal disease is **commonest in children**, but symptoms can appear late in adults.
- **Adult conditions that may warrant partial cystectomy: urachal cyst, vesicourachal diverticulum, patent urachus and urachal sinus.**
- **Communication with the bladder** (vesicourachal diverticulum, patent urachus) causes **haematuria, mucinuria or urine leakage from the umbilicus**. Infection of stagnant urine or fluid causes **fever, abdominal pain, or a periumbilical or infraumbilical fluctuant mass**; a **calculus** may form.
- **The gravest concern is malignant transformation** of remnant urachal tissue.
- **Evaluation:** most anomalies are **asymptomatic and incidental** on cross-sectional imaging. Symptomatic ones are assessed by **ultrasound or CT**; **VCUG or sinogram** shows communication with the bladder, especially with urinary symptoms or skin drainage; the **dome and anterior wall** are inspected cystoscopically for suspicious lesions.
- **Treatment is indicated for symptomatic or infected anomalies:**
  - **swab the umbilicus and send urine culture before starting antibiotics** →
  - **oral or intravenous antibiotics** →
  - **incision and drainage** of a fluctuant umbilicus →
  - **complete resolution** → **no further intervention**; **recurrent infection** → **surgical excision.**
- **Excise the entire urachus with a small cuff of the bladder dome** — including a bladder cuff ensures no residual tissue near the apex that could cause **recurrence or malignant transformation.**

### BENIGN INDICATIONS FOR PARTIAL CYSTECTOMY

| Lesion                 | Presentation                                                                            | Key investigation                             | Treatment and trap                                                                       |
|------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Leiomyoma</b>       | Women of reproductive age; obstruction, irritative symptoms, pain, haematuria           | <b>Transvaginal ultrasound, MRI</b>           | Small endovesical → <b>TUR</b> ; large/intramural → <b>wide local resection</b>          |
| <b>Paraganglioma</b>   | Episodic catecholamine symptoms; <b>syncope on micturition</b> ; or painless haematuria | <b>Plasma metanephrines + MIBG</b>            | <b>α-blockade before any instrumentation; partial cystectomy over TUR</b>                |
| <b>Endometriosis</b>   | <b>Cyclical haematuria</b> or symptoms with menses                                      | Transabdominal + transvaginal ultrasound, MRI | <b>Partial cystectomy</b> ; cystoscopy of limited use (extravesical disease)             |
| <b>Urachal anomaly</b> | Umbilical discharge, mucinuria, infected mass                                           | Ultrasound/CT; <b>VCUG or sinogram</b>        | Antibiotics ± drainage first; <b>recurrent</b> → <b>excise whole urachus + dome cuff</b> |

## Surgical Technique

- **Open partial cystectomy remains the most widely used**, given accumulated experience; laparoscopic and robotic techniques are gaining ground for benign as well as malignant disease, as a **low-morbidity alternative for lesions needing more extensive resection than endoscopy can safely achieve**.

### Open Technique

- **Lesion site decides the route: dome or anterior wall → extraperitoneal; posterior wall → intraperitoneal.**

#### *Extraperitoneal partial cystectomy*

- Opening steps as for open reimplantation — positioning, preparation, incision and **extraperitoneal bladder mobilisation**. **Pelvic lymph node dissection is usually unnecessary for benign disease**.
- With the bladder mobilised bilaterally, **intraoperative cystoscopy** confirms the site of an intravesical lesion; **intraoperative ultrasound** marks the site and extent of an extravesical or submucosal (intradetrusor) lesion.
- **Stay sutures away from the lesion**; a **longitudinal cystotomy** opens the bladder. The bladder may be partly distended as it is opened, but **the fluid must be aspirated to avoid spillage — even some benign processes recur locally**. The cystotomy is lengthened for good exposure, and the **mucosa is inspected to exclude malignancy**.
- **Circumferential transmural excision** around the lesion with electrocautery. **Even for benign disease, a margin of normal bladder is necessary** because some lesions recur after inadequate resection. **Preserve the perivesical fat** over the lesion. **If capacity allows, a 2-cm margin is recommended; biopsies of normal-appearing wall** confirm adequate resection.
- If the lesion **involves the ureteric orifice**, mobilise and **reimplant** the ureter (refluxing or nonrefluxing).
- Close the defect in **two running layers of 2-0 polyglactin** (mucosa first), then the cystotomy likewise.
- **Perivesical drain** in the retropubic space, ideally crossing the defect and cystotomy, through a stab **lateral to the rectus; 22 Fr Foley catheter**; irrigate to test; close.

#### *Transperitoneal partial cystectomy*

- Skin, rectus and transversalis fascia opened; **peritoneum entered sharply** in the midline; self-retaining retractor packs the bowel cephalad, exposing the peritoneal fold over the dome; space of Retzius and gutters developed bluntly.
- **Ureteral mobilisation is often needed**, because resection of the posterior wall may require reimplantation: incise the peritoneum over the iliac bifurcation, identify the ureter and dissect it caudally, **preserving periureteric tissue and its blood supply**.
- **Mobilise the posterior bladder** enough for resection: in **women**, incise the peritoneal fold over the **vesicouterine pouch** and develop the **pubocervical fascia**; in **men**, incise the **rectovesical pouch** peritoneum to develop **Denonvilliers fascia**, especially for lesions near the base.
- Resect as for the extraperitoneal approach.

#### *Urachal surgery (open)*

- **Periumbilical incision** carried down the midline; after the transversalis fascia, the peritoneum is identified.
- A circumferential incision around the umbilicus is taken down to peritoneum; the **peritoneal cavity is entered above the umbilicus**; with traction on the umbilical segment, the **urachus is excised with generous peritoneal wings down to the dome**.

- The bladder is distended with sterile water to allow a **transmural circumferential incision around the urachal insertion**. **A wide margin is unnecessary for benign disease**.
- Two-layer 2-0 polyglactin closure; large-bore Foley and pelvic drain; irrigate to test.
- **When the remnant is excised for infection or inflammation, the umbilicus need not be removed** in the absence of malignancy, provided subcutaneous fluid has been properly drained.

### Laparoscopic and Robotic Approaches

#### *Transperitoneal partial cystectomy*

- Positioning and ports as for **robotic diverticulectomy**.
- Mobilise the bladder off the anterior abdominal wall (incise lateral to the obliterated umbilical ligaments, divide the urachus). **The bedside assistant performs flexible cystoscopy to transilluminate the bladder** and show the lesion from the peritoneal side.
- **Posterior-wall lesions: leave the bladder suspended** from the anterior wall; cystoscopy identifies the site and the overlying peritoneum is incised directly. **Intramural or extravesical lesions** are characterised with a **laparoscopic ultrasound probe**.
- **Outline the resection** on the outer bladder surface with electrocautery to ensure adequate margins; remove the cystoscope, place a Foley, **clamp it to distend the bladder**.
- **Longitudinal cystotomy superior/anterior to the lesion**, aspirating the fluid on entry to avoid spillage; deflate the balloon; inspect the mucosa, note the lesion and its **relation to the ureteric orifices**. A **retrograde ureteral catheter** protects the ureter, or **reimplant after resection** if the ureter is at risk.
- **Full-thickness circumferential excision from inside**, preserving perivesical fat and taking an **adequate normal-mucosal margin**; specimen into a **10 or 15 cm EndoCatch bag**; **frozen sections of the margins** may be sent.
- Large-bore Foley; **two-layer 2-0 polyglactin** closure of defect and cystotomy; fill to test and reinforce with full-thickness figure-of-eight 2-0 sutures; drain through the lateral port; extract via an extended camera-port incision; close the extraction-site and 12-mm port fascia with figure-of-eight 0-polyglactin.

#### *Urachal surgery (minimally invasive)*

- **12-mm camera port 5 cm above and 2 cm to the left of the umbilicus**.
- **Unlike open surgery, the urachus is dissected out last**: the bladder is **left attached to the anterior abdominal wall** during the first part of the dissection to maximise exposure; filling the bladder with **200–300 mL** defines the anatomy.
- A small rim of bladder around the urachal insertion is scored and taken **through all layers except the mucosa**; the **entire urachus with its peritoneal wings is dissected up to the umbilicus** and divided there (umbilicus removal optional for benign disease).
- The bladder is then opened and the wall excised **circumferentially around the insertion**; specimen into an EndoCatch bag. **Frozen section of the bladder margin is unnecessary for benign disease**.
- Two-layer 2-0 closure; Foley and water test; drain via the lateral port; specimen out through an extended camera incision **or directly through the 12-mm lateral assistant port**; close fascia with figure-of-eight 0-polyglactin.

### URACHAL EXCISION — OPEN VERSUS MINIMALLY INVASIVE

|        | Open                                                           | Laparoscopic / robotic                                                             |
|--------|----------------------------------------------------------------|------------------------------------------------------------------------------------|
| Access | Periumbilical incision down the midline                        | Camera <b>5 cm above, 2 cm left</b> of umbilicus                                   |
| Order  | <b>Umbilical end first</b> , urachus followed down to the dome | <b>Urachus dissected last</b> ; bladder kept up on the abdominal wall for exposure |

|              | Open                                                              | Laparoscopic / robotic                                                                                 |
|--------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Bladder cuff | Circumferential transmural cuff; <b>no wide margin</b> for benign | Rim scored through all layers but mucosa, then opened and excised; <b>no frozen section</b> for benign |
| Umbilicus    | Preserved after drainage if no malignancy                         | Removal optional for benign disease                                                                    |

## Postoperative Care and Complications

- **Foley catheter about 1 week; cystogram before removal;** pelvic drain removed after **about 48 hours** once output falls; drain creatinine optional.
- Complications are similar for open and minimally invasive surgery and **mirror those of diverticulectomy: urinary extravasation from the resection or cystotomy site is commonest, followed by ureteric injury** — presentation and management as described for diverticulectomy.
- **Patient selection and counselling matter**, because resecting bladder wall can **reduce capacity and cause irritative symptoms** needing later intervention.
- Other complications: UTI, bladder spasm, urinary fistula, wound infection, bowel injury.

## Outcomes

- Experience is limited to **small case series**, given the rarity of these conditions in adults.
- Nevertheless, partial cystectomy — open or minimally invasive — achieves **high cure rates while preserving bladder function**.

**Concept** — *Benign partial cystectomy **borrowes the discipline of cancer surgery** — exclude malignancy, avoid spillage, take a margin — because leiomyoma, paraganglioma and endometriosis recur when incompletely excised; urachal excision for benign disease is the exception, where the whole tract matters and a wide margin does not.*

### KEY POINTS: PARTIAL CYSTECTOMY AND URACHAL SURGERY

- Confirm **adequate bladder capacity and no significant voiding symptoms** before partial cystectomy.
- Benign disease still needs **adequate resection margins**, given the infiltrative nature of many conditions and their tendency to recur after inadequate excision.
- In minimally invasive partial cystectomy, **intraoperative flexible cystoscopy by the bedside assistant** identifies the site of pathology.
- Urachal surgery for benign disease **does not require a wide margin** of dome around the urachal insertion.

## SIMPLE/SUPRATRIGONAL CYSTECTOMY

### Description

- For **neurogenic or end-stage bladder dysfunction refractory to less invasive treatment**, a **simple cystectomy with urinary diversion** may be performed; in selected cases a **diversion without cystectomy** is an alternative.
- Because cystectomy is **irreversible** and can carry **severe complications**, it is usually a **last resort** — to preserve renal function or improve quality of life.

## Evaluation and Surgical Indications

- Evaluation is multifaceted and usually follows **trials of several non-invasive treatments**: thorough history and examination, symptom assessment, **postvoid residual (PVR)**, and often urodynamics, cystoscopy and imaging.
- **It is essential that the bladder or its outlet is the cause of the problem** — patients with **bladder pain syndrome (BPS)** may continue to feel bladder pain after cystectomy.
- **Common benign indications: radiation-induced bladder dysfunction or cystitis**; neurogenic bladder from **spinal cord injury or multiple sclerosis**; refractory urinary incontinence; urinary fistula; **bladder neck contracture**; refractory **vesicourethral anastomotic stricture**.
- **In interstitial cystitis (IC)/BPS, selection is key**. Factors associated with success: a **small, fibrotic bladder**, a **small bladder capacity under anaesthesia**, and **Hunner lesions**.

## Surgical Technique

- **Principles**: mobilise the bladder, **divide the bladder pedicles**, and **amputate the bladder near the level of the trigone**. The bladder is often **opened** during the operation to identify the trigone and bladder neck. The remaining tissue at the base is **oversewn or covered with peritoneum**. **Ileal conduit diversion** is performed in the usual way.

### SIMPLE VERSUS RADICAL CYSTECTOMY

|                               | Simple (supratrigonal) cystectomy — benign                                     | Radical cystectomy — malignant |
|-------------------------------|--------------------------------------------------------------------------------|--------------------------------|
| Prostate (men)                | <b>Usually preserved</b>                                                       | Removed                        |
| Anterior vaginal wall (women) | <b>Not removed</b> unless indicated                                            | Removed                        |
| Lymphadenectomy               | <b>None</b>                                                                    | Performed                      |
| Bladder entered               | <b>Often — to find trigone and bladder neck</b>                                | Avoided                        |
| Level of excision             | <b>Amputation near the trigone</b> ; stump oversewn or covered with peritoneum | En bloc removal                |

### Open technique

- Lower midline incision → rectus fascia → space of Retzius → **peritoneum opened** → ureters identified and **disconnected** in preparation for diversion.
- The bladder may be filled through the catheter to expose the dome; the detrusor is incised and the bladder **bivalved in the midline** with a coagulating instrument — anteriorly to the **bladder neck**, posteriorly to the **cul-de-sac**.
- The lateral segments are **amputated** with the coagulating instrument; any remaining mucosa is **cauterised or excised**.
- The **posterior peritoneum** can be preserved to cover the urethral defect.

### Laparoscopic and robotic approaches

- Described, but reports are limited.
- Patient secured, pressure points padded; **supine in steep Trendelenburg**; trocars into the peritoneal cavity directed to the pelvis.
- The dome is identified and **bivalved to the bladder neck**; the pedicles are divided; mucosa is cauterised or excised. **Very friable bladder tissue may be removed piecemeal**.
- Diversion is completed **intracorporeally or extracorporeally**.

## Postoperative Care and Complications

- Care generally **mirrors radical cystectomy**, depending on the diversion.
- **Closed-suction pelvic drains** commonly stay **about 5–7 days**, removed when output is low and/or **drain creatinine matches serum creatinine**. Uncomplicated stay about **5–7 days**; many surgeons await **return of bowel function** before discharge.
- In patients with refractory benign dysfunction, simple cystectomy with diversion **improves long-term quality of life**.
- **Complication rates are substantial** and vary with comorbidity and previous treatment (major complications, graded by the Clavien–Dindo system, are common). **Commonest: wound infection, UTI, blood transfusion and ileus**.
- **Diversion without cystectomy** risks **pyocystitis, bleeding, bladder pain and secondary urothelial carcinoma**; in IC/BPS, a **substantial minority later need completion cystectomy**.
- Several reports suggest **robot-assisted surgery may reduce complications** compared with open or laparoscopic approaches.

**Concept** — Simple cystectomy is **chosen, never defaulted to**: it is irreversible, so its success depends on proving the bladder itself is the source — which is why a small, fibrotic bladder with Hunner lesions predicts success, while pain that is not truly vesical persists after the bladder has gone.

### KEY POINTS: SIMPLE CYSTECTOMY

- In **end-stage bladder dysfunction refractory to conservative measures**, simple cystectomy with urinary diversion may be offered.
- **Completion cystectomy may be required** if diversion is initially performed without cystectomy.

## BLADDER STONES, FISTULAE, AND FOREIGN-BODY REMOVAL

### Description

- Bladder calculi are **far less common than renal and ureteral stones** but can seriously disrupt quality of life.
- **The primary cause is urinary stasis**; bladder stones therefore occur mostly with **bladder outlet obstruction** — particularly BPH in men — or **neurogenic bladder**.
- Less commonly they form after **bladder substitution or augmentation with bowel**, from incomplete emptying and stasis, a **high bacterial burden**, metabolic changes in urinary composition, and possibly **increased mucus production**.
- A **foreign body** in the bladder, usually introduced **iatrogenically**, can act as a **nidus** for stone formation.

### Evaluation and Surgical Indications

- **Most bladder stones are asymptomatic** and are found during work-up of associated conditions. Otherwise **haematuria, irritative voiding, recurrent UTI, incontinence or urinary retention** prompt cystoscopy or imaging.
- Before surgery, evaluate both the **stone burden** (which decides the management strategy) and the **concomitant conditions that caused the stone** — so that recurrence can be prevented and any concomitant procedure planned.

- **Non-contrast CT gives the most reliable measurement of stone size and volume**, which guides treatment selection. Cross-sectional imaging also shows related anatomy — **prostate volume in BPH**, or **hydronephrosis in neurogenic bladder**.
- **Cystoscopy before intervention is not necessary** unless otherwise indicated — **gross haematuria**, **suspected malignancy**, or assessment of **prostatic lobar hyperplasia and prostatic urethral length**.
- **Urodynamics** if neurogenic bladder is suspected, or as part of treatment planning in BPH.

## Surgical Technique

- **Open cystolithotomy has historically been the gold standard for large bladder calculi**. Transurethral and, more recently, percutaneous endoscopic techniques have **largely replaced it**, mainly for **cosmesis and shorter catheterisation**.
- **Open surgery still suits very large stones or a high-volume burden** — maximising stone clearance and minimising operating time, because **prolonged transurethral instrumentation risks urethral stricture**.
- **Concomitant procedures often choose the route**:
  - a **very large prostate unsuitable for transurethral resection or vaporisation** → **simple prostatectomy**, ideally **transvesical**, with cystolithotomy in tandem;
  - a **large diverticulum** needing diverticulectomy to reduce stasis → cystolithotomy;
  - a **poorly compliant neurogenic bladder refractory to anticholinergics or intravesical botulinum toxin** → enterocystoplasty, with cystolithotomy at the same time.
- **Previous bladder or abdominal surgery** may alter pelvic anatomy enough to **preclude a percutaneous approach**, because of the risk of **bowel injury** at puncture and tract formation — particularly relevant with a **urinary diversion or augmented bladder**.
- **Laparoscopic and robotic cystolithotomy** offer a minimally invasive alternative **without compromising removal of a large burden**; growing experience with minimally invasive transvesical simple prostatectomy and diverticulectomy makes this attractive when **several procedures are needed at one sitting**.
- **Percutaneous cystolithotripsy** is another minimally invasive option for large calculi:
  - a **single small suprapubic incision** — better cosmesis than open surgery and **lower morbidity**, with a smaller cystotomy than open or laparoscopic surgery;
  - **shorter catheterisation and hospital stay**;
  - the **nephroscope's larger calibre and wider working channel** improve vision and allow a versatile range of instruments, giving **more efficient fragmentation and shorter operating time than transurethral cystoscopic procedures**;
  - it **virtually eliminates the risk of urethral stricture** associated with prolonged transurethral cystolitholapaxy.
- **Foreign bodies** can be removed by open surgery, laparoscopically (transperitoneal, or **transvesical LESS**) or robotically. Reported objects include **surgical materials and mesh**, **migrated intrauterine devices**, **broken sewing needles and electrical wires**.

## APPROACHES TO BLADDER STONES AND FOREIGN BODIES

| Approach                       | Best suited to | Advantage   | Limitation                                                                                      |
|--------------------------------|----------------|-------------|-------------------------------------------------------------------------------------------------|
| Transurethral cystolitholapaxy | Most stones    | No incision | <b>Prolonged instrumentation</b> → <b>urethral stricture</b> ; less efficient for large burdens |

| Approach                                                        | Best suited to                                                                                                    | Advantage                                                                                      | Limitation                                                                              |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Percutaneous cystolithotripsy</b>                            | <b>Large calculi</b>                                                                                              | Small incision, low morbidity, short catheter time and stay; <b>no urethral stricture risk</b> | <b>Prior abdominal/pelvic surgery, diversion or augmentation</b> — risk of bowel injury |
| <b>Open cystolithotomy</b>                                      | <b>Very large or high-volume burden; concomitant transvesical prostatectomy, diverticulectomy or augmentation</b> | Historical <b>gold standard</b> ; clearance and speed                                          | Incision; longer catheterisation                                                        |
| <b>Laparoscopic / robotic</b> (transperitoneal or transvesical) | Large burden with concomitant minimally invasive procedures; foreign bodies and mesh                              | Minimally invasive without sacrificing burden                                                  | Technical demand                                                                        |

### Open Technique

- Supine, **ASIS over the table break**, slight flexion; genitalia and infraumbilical abdomen prepared; **18 Fr Foley**, bladder filled with **about 200 mL** of sterile water, catheter clamped.
- **Midline infraumbilical or transverse Pfannenstiel incision** — the choice depends partly on whether a **concomitant procedure** (diverticulectomy, simple prostatectomy) is planned.
- Rectus fascia divided; recti split in the midline; space of Retzius developed bluntly (distension displaces the peritoneal reflection cephalad). **Extensive lateral mobilisation is unnecessary.**
- **Vertical cystotomy across the midline dome toward the bladder neck**, aspirating as it lengthens; **2-0 polyglactin stay sutures at the apices** prevent tearing; deep retractor blades for exposure.
- **Remove the stones (or foreign bodies) under direct vision.** Inspect the bladder for **suspicious mucosal lesions** and an **intravesical prostatic lobe** causing obstruction; perform any additional procedure for **outlet obstruction (suprapubic prostatectomy) or stasis (diverticulectomy)** now.
- Large-bore catheter; **two-layer running 2-0 polyglactin closure** — **mucosa first, then imbricating sutures burying the mucosal layer.** A suprapubic catheter is optional. Pelvic drain anterior to the cystotomy in the retropubic space on self-suction; close fascia and skin.

### Laparoscopic and Robotic Approaches

- **The transperitoneal approach is the most straightforward** for laparoscopic or robotic cystolithotomy or foreign-body excision; positioning and ports as for robotic diverticulectomy; a **fourth robotic arm is optional.**
- Mobilise the bladder off the anterior wall → **anterior vertical or transverse cystotomy** → ureteral catheterisation if needed → remove stone or excise foreign body → **two-layer 2-0 polyglactin closure** → large-bore Foley and water test → pelvic drain over the cystotomy on self-suction.
- **Transvesical multi-trocar approach (avoids the peritoneal cavity):** cystoscopy and saline filling → **5-mm trocar placed into the bladder under vision** → bladder drained → **pneumovesicum at 8–12 mm Hg** → two further trocars → stone removed or foreign body (for example eroded mesh) excised → bladder closed with absorbable suture.
- **Transvesical single-site approach:** cystoscopy and saline filling → single-site port deployed directly into the bladder → pneumovesicum → foreign body identified and excised → bladder closed; a **transurethral grasper** helps retraction.
- **da Vinci SP robot:** not described in isolation, but after transvesical access it can remove **stones, intravesical mesh and foreign bodies, or repair fistulae.** **SP transvesical stone removal at the time of simple prostatectomy and vesicovaginal fistula repair** have been described.

### Percutaneous Technique

- **Dorsal lithotomy or supine;** perineum and lower abdomen prepared and draped, ideally with a **fluid-control pouch**.
- **Sequence:**
  - transurethral Foley; **bladder distended with saline** →
  - **1-cm skin incision about two fingerbreadths above the pubic symphysis** →
  - **18-gauge trocar needle** into the bladder; aspirate to confirm — under **ultrasound guidance** or with **flexible cystoscopic vision**. After previous abdominal surgery, use **preoperative cross-sectional imaging or real-time ultrasound to exclude interposed bowel** →
  - remove the inner obturator; **return of urine confirms bladder entry** →
  - pass an **Amplatz Super Stiff PTFE-coated 0.038-inch guidewire**, coiling it in the bladder; remove the trocar →
  - **Amplatz dilators sequentially to 30 Fr**; place a **30 Fr sheath** across the tract →
  - insert a **24 or 26 Fr nephroscope**; the sheath's wider calibre allows **continuous irrigation** and clear vision.
- **Stones under 1 cm:** retrieve intact with **rigid grasping forceps**. **Larger stones:** fragment with a **pneumatic or ultrasonic lithotripter** under vision, then retrieve fragments with forceps.
- When cleared, pass a **suprapubic catheter over the guidewire** to gravity drainage, or leave a **large-bore urethral Foley** instead.
- **Entrapment-bag technique:** a laparoscopic entrapment bag is passed through the 30 Fr sheath; with a **flexible cystoscope** the stone is captured and the bag cinched and exteriorised to skin level (sheath removed); the bag is opened, the sheath reintroduced **into the bag**, and the stone fragmented inside it with nephroscope and lithotripter; the bag is then pulled out across the tract, and a suprapubic catheter placed over the retained guidewire.

### Postoperative Care and Complications

- **Foley catheter about 1 week** for healing of the closure; **pelvic drain** monitored — **high output may indicate a leak**; removed at about **48 hours** if output is low and catheter drainage adequate. **Drain creatinine is not needed unless output is high**. **Cystogram before catheter removal**.
- **After percutaneous cystolithotripsy with a suprapubic tube:** **clamp the tube for 24–48 hours for a trial of voiding**, and remove it once the patient voids. A **urethral catheter**, if used instead, is removed **24–48 hours** after the procedure for a trial of voiding.

**Concept** — *Stone surgery treats **the stone and the stasis together**: stone burden chooses the route, but the operation needed for the cause — prostatectomy, diverticulectomy, augmentation — often chooses it first, and prior abdominal surgery can close the percutaneous door.*

#### KEY POINTS: BLADDER STONE SURGERY

- The approach depends on the **size and number of calculi** and on any **associated process causing stone formation** that must be addressed at the same time.
- **Percutaneous cystolithotripsy** is a minimally invasive alternative for **large calculi**, but percutaneous access **may be prohibited after prior pelvic or abdominal surgery** because of the risk of bowel injury.